The Reliability Stack: A Cross-Layer Taxonomy for Engineering Reliable AI Systems

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Abstract

Reliability engineering for AI systems is dominated by two narratives: scale a general-purpose model, or hand-assemble safeguards per problem. We ask the routing question between them: when does scaling buy reliability, and when do changes at other layers—task and risk specification, learning objective, representation, inference architecture, output contract, and assurance—buy more? We organize the reliability-bearing decisions into a cross-layer taxonomy, the Reliability Stack (L0–L5), and read reliability claims as inherited failure envelopes per layer rather than per-method scores (the objective space is still expanding: hierarchical code prediction and reasoning post-training entered the map while we drafted [89]). From this we derive a selection framework: two axes, answer-space openness and cost of error, return a candidate operating mode plus the evidence a team needs before deploying it, with thresholds set by a stated risk budget. The synthesis is evidence-typed by construction: eighteen claim families, each carrying one of six evidence classes ([T] theory, [E] independent empirical, [A] author-reported, [V] vendor-reported, [S] verified synthesis, [W] watch), applied to individual numbers rather than whole papers (Fig. 25), with three contested rows named — calibration direction, language-JEPA stability, monitor-evasion attackability — and exclusions recorded. Across seven stack-instance case studies—grounded assistant, decision-model serving, hybrid specialist, latent-prediction planner, discriminative baseline, a rules/formal-verification null, and production diffusion serving—no reliability currency is guaranteed at any single layer: scaling buys coverage and fluency, contracts buy validity, assurance buys operational dependability. The engineering problem is choosing which currency the task needs and which layer to change next.

1 Introduction

1.1 The motivating question

A settled habit in AI engineering treats progress as trajectory: make the general-purpose model larger and better-trained, and reliability follows. The habit deserves the question, and it rarely gets asked in the form engineering decisions need: *when does scaling a general-purpose model improve reliability, and when does changing how a system learns, represents, decides, or verifies buy more of it?* The question admits documented wins on both sides, and the engineering-relevant answer is

*Generative AI tools were used for research assistance, drafting, and editing. The author reviewed the manuscript and takes responsibility for its claims, citations, and artifacts.

a *routing rule* [136]: given a task's openness and its error cost, which stack layer should absorb the next round of engineering work?

1.2 What we observe

The parts exist but are not connected. Objective-space research knows what each training signal cannot deliver [12, 87, 118]. Deployment-facing surveys catalogue failure modes and safeguards [66, 117, 128]. Benchmark and instrument efforts assess calibration, abstention, and interface stability [95, 114]. The assurance literature owns a mature toolset: conformal and provable-alignment selective prediction, monitoring, human escalation [39, 150]. The connective tissue is missing: no survey we examined integrates the stack's layers, from learning objectives and representations through architectures and output contracts to decision interfaces and assurance, into one *selection* procedure for reliability-constrained workloads (Section 12). [1] Model-family surveys answer a different question (which generative model family exists and how it trains [3, 155]).

1.3 What we contribute

- **C1 — the Reliability Stack (L0–L5):** a cross-layer taxonomy that assigns each reliability decision to one of six named layers (task and risk specification, learning objective, representation, inference architecture, output contract, assurance and control), with evaluation/monitoring cutting across all layers (Fig. 1). The objective-space map is maintained against a moving field (hierarchical next-scale code prediction and reasoning post-training both entered it while we drafted [89]), and the taxonomy is framed by what each layer *cannot* deliver.
- **C2 — a selection framework** that consumes the taxonomy: routing by answer-space openness and cost of error, four placement modifiers, a six-step selection procedure, thresholds derived from the risk budget, and escalation conditions instead of "safe zones" (Fig. 24 and Section 10). It is exercised on seven author-chosen stack instances and on a four-incident retrospective probe (Appendix A); the probe's cases were chosen for record quality, which is also what makes their design choices visible, so it tests applicability, not generalization.
- **C3 — an evidence-typed synthesis:** a claim ledger that separates theory-under-assumptions, independent empirical results, vendor-reported claims and synthesis, with an exclusions register and contested rows that are surfaced rather than footnoted (Section 11 and Fig. 25).

None of C1–C3 is a new model or mechanism. The contribution is the argument plus the ledger: reliability is an assembly property, and the assembly can be selected with instruments that already exist — calibration, contracts, monitors, escalation.

1.4 Paper map

Section 2 fixes the reliability vocabulary, and Section 3 introduces the stack. Sections 4–8 walk the layers L1–L5 with method cards and per-family envelopes; Section 9 develops seven stack instances with their inherited failure modes and named evidence gaps; Section 10 presents the selection framework; Section 11 states the evidence typology used throughout; Section 12 positions the work against the three genres of prior work; Section 13 discusses where the stack itself fails, eight named measurement gaps, and the limitations of survey-grade evidence.

[1] Comparison set: Section 12, corpus cutoff 2026-10-17. The works there do not integrate the five positioning axes into one selection procedure: the closest cross-layer survey [145] is a roadmap of threats, the closest selection procedure [6] is scoped to inference-time scaling, and the January 2026 five-axis survey of 147 systems [37] uses different axes (architecture, memory, planning, coordination, safety).

2 Reliability, Precisely

Throughout this paper we use the dimensional vocabulary below, and each dimension is chosen to vary independently: matching on one dimension tells you nothing about the others, and improving one does not require changing the rest.

2.1 What reliability is

Operationally, a configuration is reliable only relative to three named terms: an envelope of failure kinds it must not produce, a budget it must hold, and an assurance path that intervenes as it drifts. "Reliable" carries those relative terms; the selection procedure of Section 10 is how they are chosen. Reliability is not benchmark accuracy or fluency [16, 83]. A system can raise benchmark scores while becoming less deployable, more confidently wrong, less able to refuse, more expensive to operate within a budget [136]. Reliability is a property of the *system configuration* as well: the contract, thresholds, monitors and escalation paths below belong to that configuration, not to a single weight checkpoint. That is why our unit of analysis per Section 3 is the assembled stack instance.

Classical reliability engineering supplies the vocabulary this section borrows, and three distinctions do most of the work. Reliability is the continuity of correct service and availability is readiness for correct service [13]; safety is freedom from losses to people and environment, a property of the controlled system [93]. In deployment terms we read availability as uptime, which can be high while outputs are wrong; that gap is why we keep the two apart [J]. Second, reliability is compositional: a chain of redundant components is only as independent as its failure causes allow, and common-cause failures defeat redundancy that looks adequate on paper, which is why a stack that adds a monitor and a verifier has added no independence unless the model, the monitor, and the verifier fail for different reasons [93, 94]. Third, accident analysis separates latent conditions, present in design and organization long before an event, from active failures at the point of operation [93]. Our inherited failure envelope is the stack's latent-condition accounting: the failure a layer defers is a latent condition for the layer below. Conventional analysis grades each failure mode by severity, occurrence and detectability (FMECA, IEC 60812 [79]); we use it to rank layers by what they leave unrecoverable; hazard frequency is a different axis. Of these terms, only *reliability debt* is a coinage; the envelope is our adaptation of latent-condition accounting.

Every claim in this paper carries one of six evidence classes: [T] theory under stated assumptions, [E] independent empirical, [A] author-reported, [V] vendor-reported, [S] verified synthesis, [W] watch/pending. Four annotations record what a claim is reacting to, while the class carries the strength of the evidence: [C] contradiction, [N] named gap, [GAP] unmeasured link, [J] our analytic judgement. Table 3 is the single alphabet every other surface cites, and Section 11 gives the full protocol.

2.2 The dimensions

Inherited failure envelope. The set of failure modes a layer's choice makes unrecoverable downstream: the choice cannot remove them, it can only relocate them. When a fix moves a failure to another layer rather than removing it, we call that *reliability debt*.

Well-posedness under distribution shift. A task is well-posed when its answer space and input distribution can both be stated; a model is reliable only relative to that pair. Deployment drifts the second term, which is why shift is a first-class modifier in Section 10, and why the classifier-baseline literature, with the longest experience of this instability, anchors the dimension [61, 103].

Calibration and selective prediction. Calibration is measured confidence–accuracy agreement. Selective prediction turns that agreement into a decision by fixing a cutoff with a stated error rate [39, 148]. The formulation is older than the LLM literature: fix a cut on a confidence score, accept the region it selects, and report the error on the accepted set [59, 60]. What changes in the LLM setting is the calibration error of the score, confidence that need not be monotone across an option set, and a rejected sample that has already cost a generation. The direction of miscalibration (over- vs. under-confidence) changes which way a confidence-gated cascade fails [114]; we therefore track direction as well as magnitude.

Validity versus truth. Syntactic/schema correctness of an output is separable from semantic correctness of its content [40, 63]. Conflating these is the source of most spurious "guaranteed" deployment claims, and the asymmetry shapes Layer 4 (Section 7).

Grounding. For generative output, whether a claim is supported by available evidence. The trustworthy-RAG literature treats grounding as a collection of techniques organized by dimension, each opening new failure paths of its own (stale sources, retrieval misses, unfaithful use) [117, 162].

Error cost and reversibility. Errors differ by consequence severity and by whether they can be undone. This becomes the L0 risk budget: the escalation band of Fig. 24 is derived from it, and the band's width is our answer to how much error a task can absorb.

Prediction unit and generation schedule. What a system predicts — a token, a continuous latent, or a discrete code spanning several tokens — and the order in which predictions condition on each other are separable from both the training signal and the architecture: the same objective admits a left-to-right chain, an iterative denoising sweep, or a coarse-to-fine level order (Fig. 11). The schedule fixes where commitment points fall, and with them whether early commitments remain revisable.

Latency, cost and budget. Deployment reliability is bounded by operational budget; a configuration that is reliable at unit cost can be unreliable at required throughput. We treat budget as a selection constraint; maximizing it is a different objective.

Execution configuration. The inference engine, the numeric precision, the cache settings and the co-resident traffic a model is served under: properties of the deployment; the weights alone do not carry them. They enter the computed function, so an error rate belongs to the configuration it was measured on (Section 8.7).

Monitoring and drift response. This dimension is the assurance layer's own reliability: monitors can be evaded under ordinary task pressure [131], and OOD detection alone does not close the failure envelope [66].

2.3 Three determinism terms: task, model, serving path

Section 10 routes tasks on two axes: how open the answer space is, and what an error costs. That routing requires one distinction we fix here. A *task* has a well-defined or an open-ended answer space; a *model* either samples stochastically or is effectively deterministic at inference; and a *serving path* either reproduces the same arithmetic for the same request or does not, because the inference engine, the numeric precision and the co-resident traffic on the device all enter the computed function. The first two are frequently conflated in vendor material ("deterministic mode" versus "deterministic task"), and we keep them strictly separated. The third is the one a deployment team discovers by accident: no checkpoint records it, so a threshold fitted on one serving path does

not carry to another (Section 8.7). The planner on a well-posed route query and the generative assistant on an ambiguous brief face different reliability demands even under identical decoding parameters.

2.4 Vocabulary used in this paper

Layer. One of the reliability-bearing decision problems the stack arranges, from task and risk specification to assurance and control; a decision problem, not a model class.

Stack instance. A particular filling of the layers by one assembled system; the unit is not a model considered on its own.

Inherited failure envelope. The failure modes a layer's choice makes unrecoverable downstream: the choice cannot remove them, only relocate them.

Reliability debt. What a fix incurs when it moves a failure to another layer rather than removing it; see also *inherited failure envelope*.

Well-posedness. A task is well-posed when its answer space and input distribution can both be stated; reliability is claimed only relative to that pair.

Selective prediction. Turning measured confidence into a decision by fixing a cutoff with a stated error rate.

Validity versus truth. Schema or syntactic correctness of an output, which a contract layer can enforce, versus semantic correctness of its content, which no contract can.

Residual. The error left after the safeguards act, which the operator chooses to own and re-evaluate rather than assume absent.

3 The Reliability Stack

The Reliability Stack (L0–L5) arranges reliability-bearing engineering decisions in one frame — task and risk specification, learning objective, representation, inference architecture, output contract, assurance and control. It is a stack of *decision problems*: a fix at one layer can move a failure to another rather than remove it — we call that reliability debt. An objective that buys generative quality does not buy an assurance story, and a large label set does not make a typed decision calibrated.

Six archetypes recur in practice: generative LLM, discriminative classifier, decision-oriented model, hybrid specialist, JEPA world model, and a rules/verification pipeline (the last carried as an *honest null*: we include it because no verified production instance exists, and that absence is itself the finding [N]). Each is a *stack instance* — a particular filling of L0–L5 — and they differ at identifiable layers (Fig. 1). Case 1 carries the generative and grounded halves; Case 6 the verification half, and composing the two is a stack instance rather than a seventh archetype. Case 7 adds a second instance of the generative archetype, under the factorization that Table 1 row 3 names, so six archetypes stay six. Reading them as orthogonal rival model classes is the misreading the case studies of Section 9 are built to prevent.

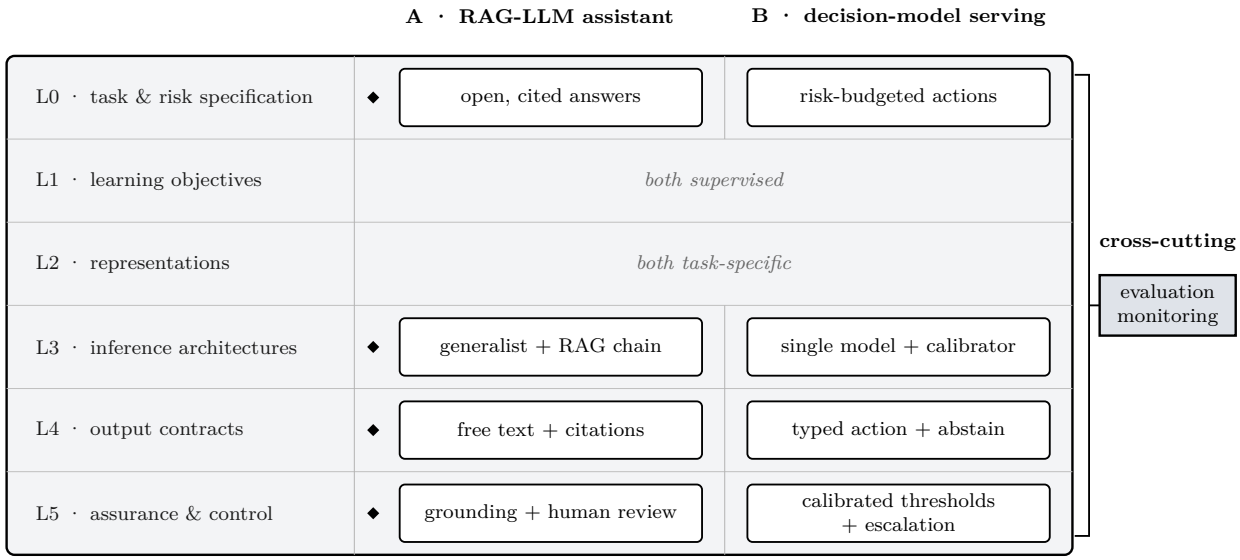


Figure 1: The Reliability Stack (L0–L5) with two example instances in parallel columns. Layers read top-down, L0 specification to L5 assurance; a choice at any layer constrains what the layers below can deliver. Cells appear only where the two instances choose a different kind of thing, and the diamond marks those four layers; L1–L2 carry one shared phrase because both instances make the same kind of choice there. No layer repairs another's gap: the layers assemble into a stack, and no layer outranks another. Layer cuts and cell placements are an analytic synthesis [J]; instance cells are candidate placements — column B vendor-reported [V], column A our own reconstructed instance [S]; evaluation + monitoring apply at every layer.

4 Layer 1 — Learning Objectives

This layer collects the method cards for *how a system is trained*; the shared objective-space map is Table 1, and the four-signal comparison is Fig. 2.

4.1 An objective-space map: Table 1

The first layer of the Reliability Stack asks what a model is trained to be. We treat a *learning objective* as the pair (training signal, probability factorization): autoregressive next-token prediction combines a factorization of the joint distribution with a negative-log-likelihood signal [69, 82, 140]; a language model's architecture (mostly Transformer) is a separable choice. Keeping the two apart matters for reliability claims, because a change of factorization converts an architectural family into a different objective [57], and a change of architecture under a fixed objective changes failure modes without changing what the signal encourages.

Table 1 maps the ten objective families our review identifies (the two newest entered while we were drafting: hierarchical code prediction, included because it moves failure into the tokenizer rather than removing it, and reasoning post-training). The columns follow the method-card schema of Section 11: the modelling principle, the reliability question it inherits, and — critically — the properties the objective does *not* deliver. Our claim throughout is that a strong score on any of these signals is not a downstream property: likelihood, fidelity, sample quality, calibration and

Family	Modelling principle	Reliability question	Not guaranteed / fails	Ev.
Autoregressive NLL	next-token conditional prediction	does sequence mastery buy factual + constraint correctness?	fluency ≠ grounding; poor calibration; unbounded generation	[E]
Masked / denoising	corrupt-then-recover	does recovery training build faithful representations?	generation is not the objective; encoder-only scope	[E]
Diffusion / flow matching, incl. production serving (§4.2)	iterative denoising, learned vector fields	quality–latency–risk trade-off of the sampler?	multi-step cost; variability across samplers; risk is reference-relative	[T, E]
Energy-based	inference as energy-landscape search (NRGPT)	does a different inference principle change reliability?	small-scale evidence; no superiority vs. production AR	[T, E]
Supervised discrimination	$P(y \mid x)$ mapping head	is the mapping stable in deployment?	distribution shift; no "why"; label-space drift	[E]
Contrastive (CLIP-class) [125]	agreement of matching pairs	does embedding similarity imply semantic correctness?	retrieval errors; OOD embeddings; fine-grained mismatch	[E]
Latent prediction (JEPA family)	predict target representations in latent space	which structure is recoverable, and does prediction help planning?	*JEPA paradox in language*: authors report conditional-concentration failures on ambiguous text	[A]
Hierarchical code prediction (NCM-class)	coarse-to-fine multiscale residual VQ codes; all codes per level in parallel	does the tokenizer carry meaning exact enough for downstream checks?	tokenizer-stage substitutions (entities, numerals); codebook compounding: authors' provisional analysis; no released production guard	[A]
Preference / reward post-training	alignment training on top of an objective	does the reward measure what we want?	reward hacking [55]; miscalibrated verifiers [127]	[E]
Reasoning post-training (RLVR)	policy optimization against a programmatic checker, reusing the AR factorization	does verifier-checkable correctness at training time survive deployment?	verifier gaming [127], length inflation [50], coverage loss at large k [157], unfaithful traces [139], compute saturation [153]	[E, A]

Table 1: Objective-space map. Each family is optimized for a different currency; no column is interchangeable. Ev. = the class of a row's strongest anchor; where two classes appear, the first governs. Chips carry the dominant class of the single claim they name; [C]/[N] mark contradiction and absence; the full alphabet is Table 3. Rows are *candidates*; the selection procedure lives in Section 10.

semantic correctness are different currencies [31, 118, 152].

Three observations from Table 1 carry through the paper.

(i) Transfer: an objective that works in one modality need not concentrate its signal for another. The clearest documented case is the JEPA paradox: deterministic squared-error latent prediction presumes *conditional concentration* of target representations, which masked text violates because several valid continuations are near-equidistant [87, 158]. The same caution applies to importing image-era diffusion assumptions into language: D3PM ties masked-diffusion training

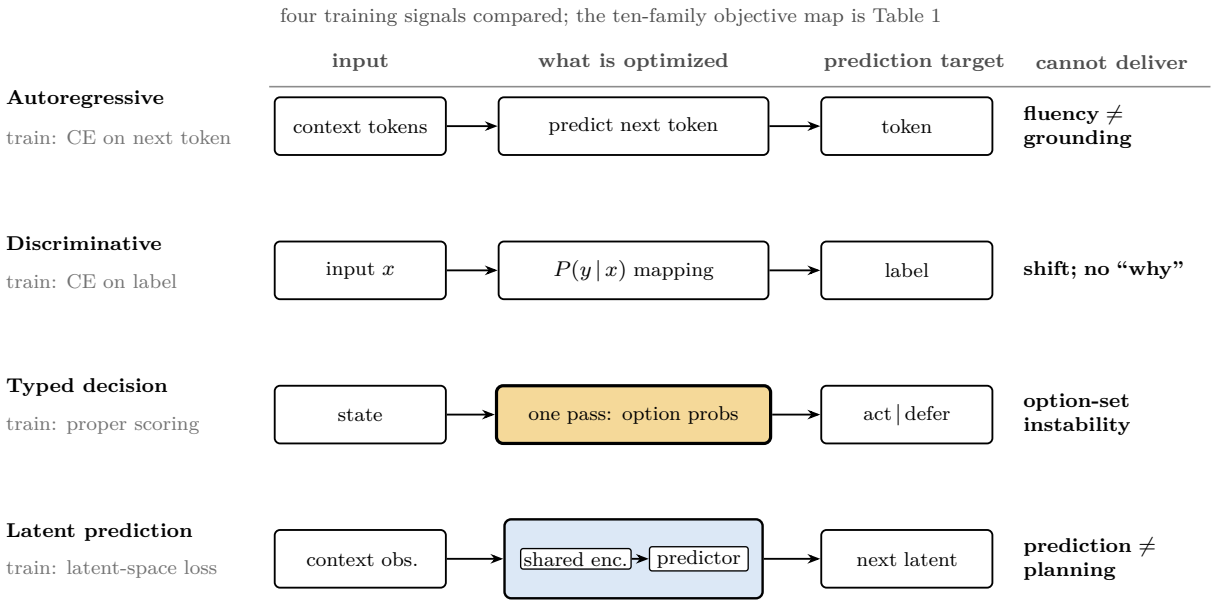


Figure 2: Four training signals on one stage grid — input, what is optimized, prediction target, and what the signal cannot deliver: autoregressive next-token, supervised discrimination, typed decisions, latent prediction. The amber chip marks the typed-decision mechanism, the item the ledger carries as threshold-as-policy [V] with independent benchmarking [E] (Section 7); it is a claim, not a failure; each row's failure sits in the last column. The grid compares four signals; the ten-family objective map is Table 1. Signals and their failures are candidates.

to discrete state spaces [12], recent theory shows those trainers implicitly learn token-order strategies, i.e. a factorization in disguise [57], and controlled generation arrives only with a redesigned continuous objective [98].

(ii) Historical trade-offs are our thesis in miniature. GANs optimized realism via adversarial play and inherited mode collapse and unstable games [8, 31, 64]; VAE regularized a latent space and paid with posterior collapse and blurry samples [67, 86]; normalizing flows bought exact likelihood with architectural constraints on capacity [120]; flow matching replaced stochastic diffusion transport with regression along probability paths, separately from any invertibility constraint [102]. Each is an instance of our central question — *the objective decides what the system cannot deliver* — long before the scale debate; we use them as comparative rows (the catalogue genre is already served, e.g. [3, 155]).

(iii) The assurance layer is objective-agnostic and always needed. Grounding, validators, calibrated acceptance and escalation (L5) wrap *any* row of Table 1; the objective changes only *where* the inherited failures sit; assurance is needed on every row. The block-diffusion serving literature is today's cleanest demonstration that an inference configuration can be chosen against a reference-relative risk budget rather than a mean benchmark score, and that the same selection procedure transfers, up to precision limits and with batch-invariant sampling, to speculative-decoding acceptance rules and to weight quantization [92] — the pattern we generalize in Sections 7–8.

4.2 Family cards: objectives, envelopes, complements

We expand the ten Table 1 rows into eight compact cards: what the objective *optimizes*, where it *excels*, where it *fails*, and which complementary chapters (Sections 7–8) cover the residual. Eight

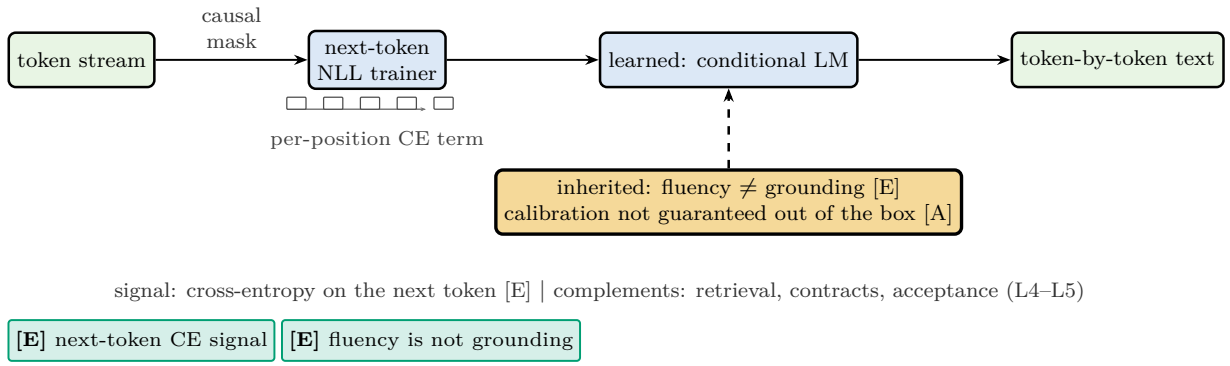


Figure 3: Family card: autoregressive next-token prediction. Yellow = the inherited failure with the strongest ledger support.

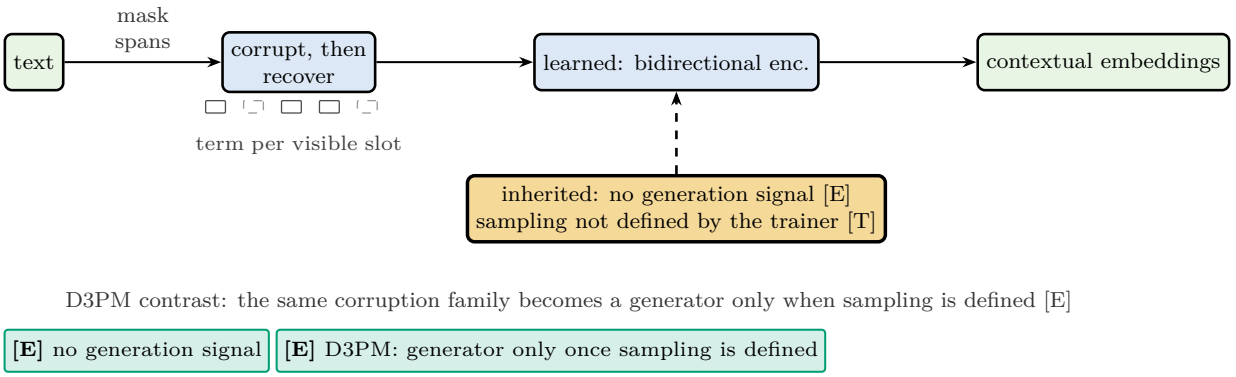


Figure 4: Family card: masked / denoising objectives.

cards cover ten rows because discrimination and contrastive learning share one card — their inherited failure is the same mapping-stability question (Fig. 7) — and latent prediction (JEPA) is argued in prose below and instantiated in Case 4 (Section 9.4), which is where its envelope becomes measurable. Each card reads left to right — data in, the training mechanism, the learned object, the output — with the inherited failure hung below the learned object on a dashed arrow and the training signal in the footer. Cards are candidates; the selection procedure in Section 10 consumes them.

Autoregressive next-token prediction. The mechanism is sketched in Fig. 3.

Optimizes conditional sequence likelihood (left-to-right factorization over the joint, NLL signal); compute scaling follows observed laws [69, 82]. *Excels at* open-ended generation, in-context adaptation, and tool-readable instruction following — the substrate every layer above assumes. *Fails at* factual guarantees (fluency is not truth; hallucination surveys report persistent inconsistency across elicitation formats [2]) and calibration out of the box. *Complements:* retrieval grounding, output contracts, calibrated acceptance and escalation (§7–§8).

Masked and denoising objectives. The mechanism is sketched in Fig. 4.

Optimizes recovery from corruptions; trains representations that transfer broadly. *Excels at* encoder tasks and representation learning with modest data. *Fails at* generation: the training signal is recovery, and the D3PM contrast with masked-diffusion generation shows the same corruption

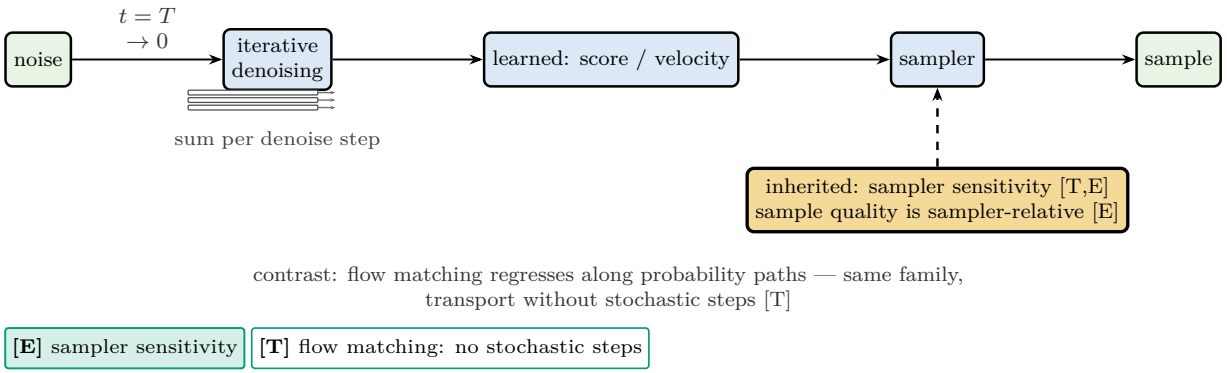


Figure 5: Family card: diffusion and flow matching (sampler sensitivity as the inherited failure).

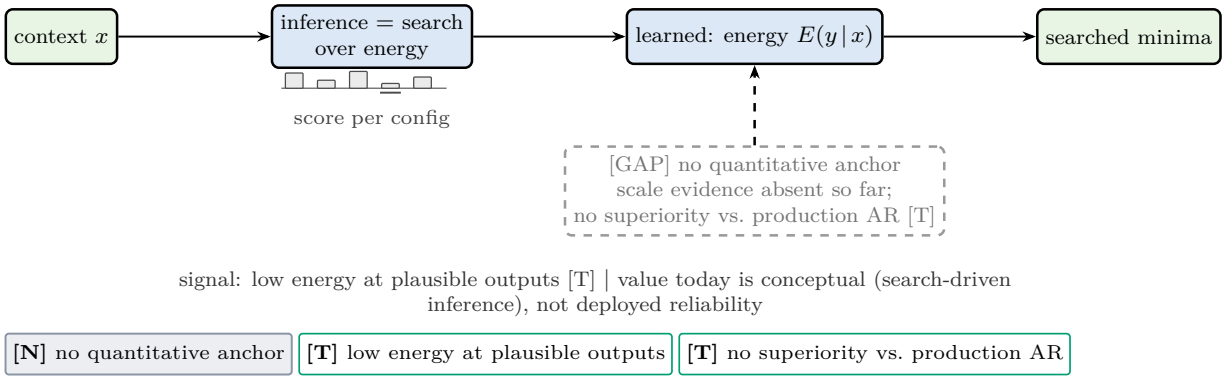


Figure 6: Family card: energy-based objectives. Gray = absent evidence ([N]): no quantitative anchor in our ledger; the card's [GAP] label marks the specific unmeasured link behind that absence, not a second finding.

family splits into representation-trainers and sequence-generators only when sampling is defined [12, 57]. *Complements:* pair with a decoding layer or use as the base for specialist encoders.

Diffusion and flow matching. The mechanism is sketched in Fig. 5.

Optimizes iterative denoising (DDPM; improved variants tie likelihood to FID-style diversity [68, 118]) or regression along probability paths [102]; discrete state spaces and controllable formulations carry these into text [12, 98], with parallel-decoding latency arguments spanning the DLM survey [97], Turkish masked-diffusion proof-of-concept Diffutron [164], and order-learning bridges [57]. *Fails at* sampler sensitivity: the same trainer yields different samples across samplers (DDIM [137]); the comprehensive survey documents likelihood-vs-sample-quality separations [152]. *Complements:* risk-budgeted serving (§8.5) turns sampler choice into an assurance question rather than a benchmark answer.

Energy-based objectives. The mechanism is sketched in Fig. 6.

Optimizes an energy landscape whose minima are plausible outputs (NRGPT class [34], [T]); inference is an iterative search. *Excels at* making inference a tunable computation, in principle aligning effort with difficulty. *Fails at* scale evidence: demonstrations remain small; no superiority over production AR systems is established, and we flag this as an honest open row rather than a case for exclusion. *Complements:* the usual assurance stack applies; value today is conceptual

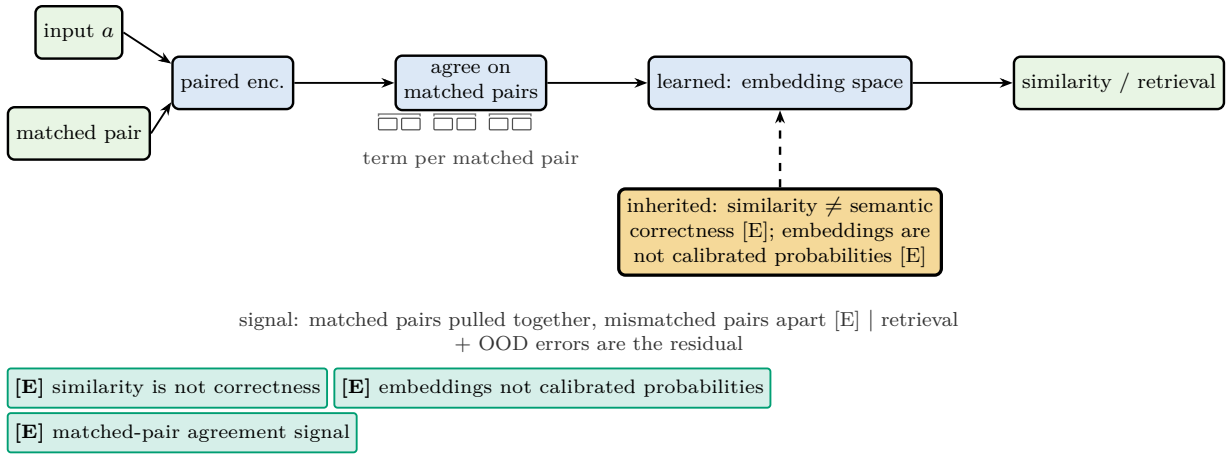


Figure 7: Family card: contrastive representation objectives (Table 1 rows 5–6 — the supervised-discrimination family shares this row's computed-agreement signal; its own row lives in Table 1).

(search-driven inference).

Discriminative and contrastive representation objectives. The mechanism is sketched in Fig. 7.

Optimizes a mapping head or agreement of pairs; typically the cheapest, most auditable family. *Excels at* narrow, well-instrumented decision regimes where OOD -monitoring and calibration methods have matured (§8 cites the UQ surveys [103, 148]). *Fails at* open-ended tasks and distribution shift; embeddings are not calibrated probabilities and break threshold policies built on them. *Complements:* acceptance/escalation machinery, designed jointly with the L4 contract.

Latent-predictive joint-embedding objectives (JEPA family). *Optimizes* predictability of target representations under partial context (I-JEPA lineage; the identifiability gap we treat as open in Section 5.2 [87]). *Excels at* vision-style regimes where the objective matches the data's structure; attractive because prediction concentrates signal without generative cost. *Fails at* language: the paradox thesis is that ambiguous-text targets lack concentration, so squared-error training degrades [158]; assumption conditions (e.g. non-Gaussian identifiability lifts [38]) remain unclosed in production settings. *Complements:* world-model planners need an output contract (§7) and monitors (§8) exactly because prediction quality alone does not certify planning quality.

Preference, reward and rule-grounded post-training. The mechanism is sketched in Fig. 8.

Optimizes alignment of outputs to explicit preferences or verifiable rules on top of any base objective (RLHF/DPO families; the Laya-class recipe trains decision heads with proper scoring rules [V]; the Jev-class System-One benchmark family shows decision-model serving gains [35]). *Excels at* instilling response shape and typed decision distributions without re-architecting. *Fails at* reward hacking and miscalibrated verifiers: the signal reflects the reward model's beliefs, not ground truth [55, 127]; direction-of-miscalibration disagreements across independent reproductions are documented for the Laya-class architecture and reported as contested and unresolved (§7). *Complements:* The entire L5 stack; post-training changes the prior, assurance changes the posterior of use.

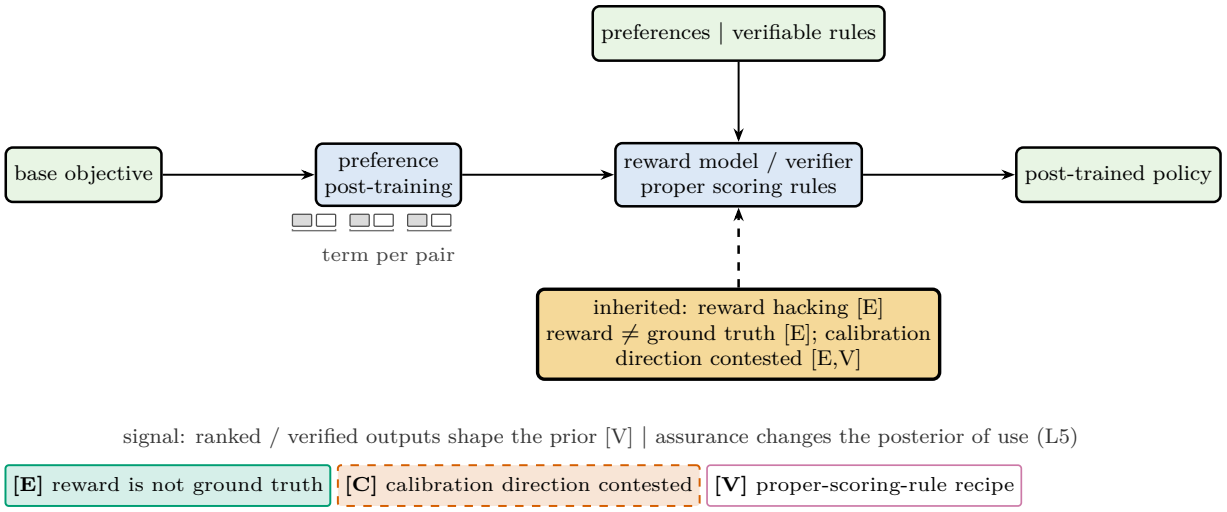


Figure 8: Family card: preference / reward post-training (contested calibration direction named).

Reasoning post-training (RLVR). The mechanism is sketched in Fig. 9.

Optimizes expected verifiable reward over a rollout group: a programmatic checker scores sampled chains under KL/clip, keeping the autoregressive factorization and changing only the signal. *Excels at* domains with executable ground truth, where it converts latent capability into pass@1 — R1-Zero moves AIME 2024 from 15.6 to 77.9 [33]. *Fails at* the checker: fuzzed verifiers return false-positive rates of 0.832, 0.869 and 0.557 by verifier family [127]; on Qwen2.5-Math-7B/MATH-500, randomly assigned rewards still buy 21.4 pp of gain against 29.1 pp from ground truth, though Llama3 and OLMo2 do not reproduce the gain [133]; base models overtake their RLVR versions at large k [157]; length inflation follows from the loss on incorrect answers [50]; returns diminish polynomially [153]; and traces need not be faithful [139]. *Complements:* typed outputs make rewards checkable (§7); assurance audits the checker as well as the policy.

Hierarchical code prediction (NCM-class). *Optimizes* a coarse-to-fine factorization over levels with a residual-VQ code signal: a frozen multiscale residual VQ-VAE tokenizer represents text as K levels of discrete codes over L tokens (with ℓ_k the number of code positions at level k, $K \ll L$), and a transformer generates each level in one parallel forward pass conditioned on the cumulative reconstruction — inheriting next-scale prediction from image hierarchies [89]. *Excels at* short generation paths ($K \ll L$): reports 21× faster decoding on LM1B at matched parameter scale and 31× on XSum at matched ROUGE-L [A], with quality at or above masked-diffusion baselines, Sudoku accuracy above all evaluated baselines [A], and 4.9× lower training-token exposure [A] [89]. *Fails at* tokenizer-stage semantic risk: reconstruction is near-exact per dataset (98.79–100% [A]), but the released evaluation shows substitutions of named entities or numerals — a 99.79%-reconstruction summarisation case [A] does not certify meaning-level fidelity — and the tokenizer fixes what downstream checks can distinguish. Math/data/correctness goals shift the exact-recovery requirement, and the authors' conditioning-corruption training buys code-error robustness inside the representation, not at the contract ([A]-class). *Complements:* output contracts must define token-exactness vs. meaning-equivalence boundaries; acceptance machinery resumes.

A schedule is a factorization choice, not an architecture, and the objective families above instantiate three of them (Fig. 11). Autoregressive generation is the chain rule: one commitment per token, none revisable. Iterative denoising revisits every position on every pass, so revision is free

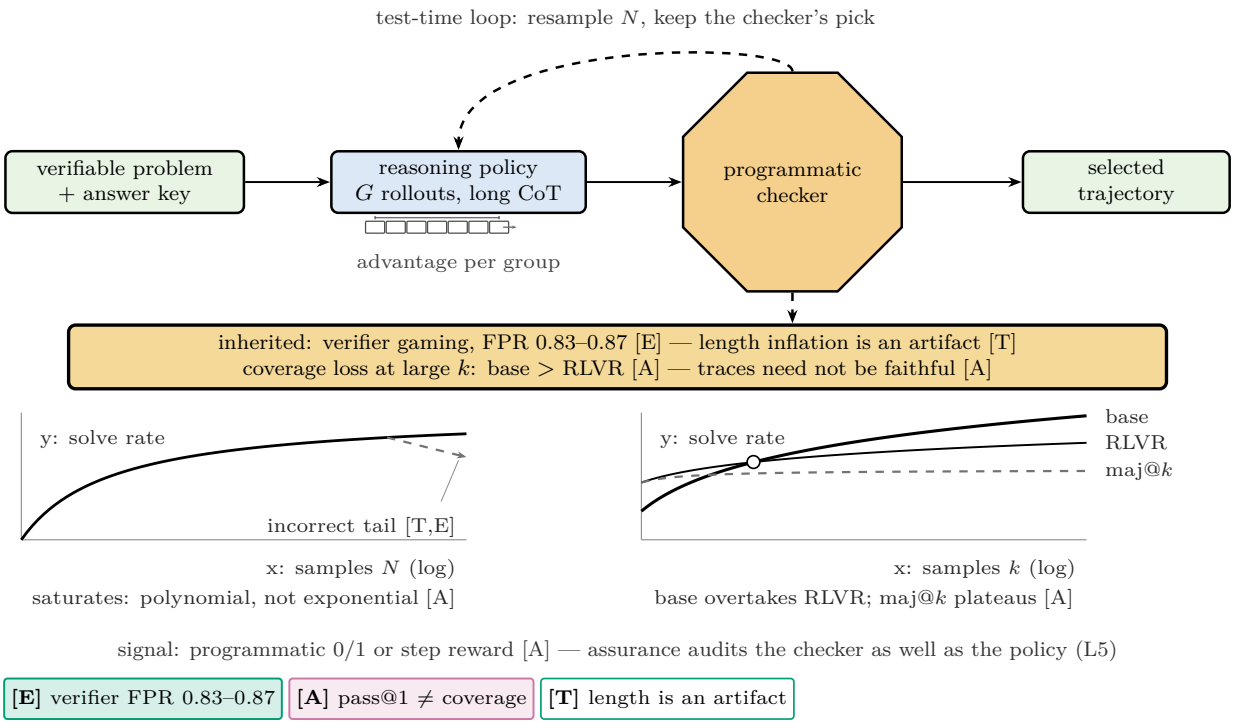


Figure 9: Family card: reasoning post-training (verifier gaming as the inherited failure).

but nothing is settled until the last pass. Coarse-to-fine hierarchies commit one level at a time and stay revisable until the next level resolves them, at $K \ll L$ sequential steps. What differs across the three is how many sequential decisions an output contract must accept and whether any of them can be taken back — the quantity a contract inspects at acceptance time (Section 7).

What the cards jointly establish. No family wins across currencies: likelihood, fidelity, calibration and semantic correctness are different objectives, so a stack that mixes families (L1) with contracts (L4) and assurance (L5) is the actual engineering answer, and Section 10 formalizes the choice.

5 Layer 2 — Representations and What They Guarantee

5.1 Representation as the interface between objective and system

The first layer decides what a model is trained to be (Section 4), and the second decides what the input becomes; a *representation* is the interface between the two. It fixes the objects the system computes over, and with them the assumptions that every later layer (inference architecture, output contract, assurance) inherits. Three families recur across the stack instances of Section 3. (i) *Token embeddings*: a discrete sequence of subword units, represented as learned vectors. This is the dominant interface of language systems, and it embeds a modelling decision that is easy to overlook. Tokenization is more than compression: comparative studies find that tokenization choices impose inductive biases of their own, with downstream consequences that compression ratio does not predict [130]. [E] Byte-level and multiscale byte-sequence designs (the ByT5/MEGABYTE class) relax or remove the tokenizer at the cost of longer effective sequences, though multiscale byte transformers were designed to amortize that cost [156]. (ii) *Encoder and joint-embedding latents*:

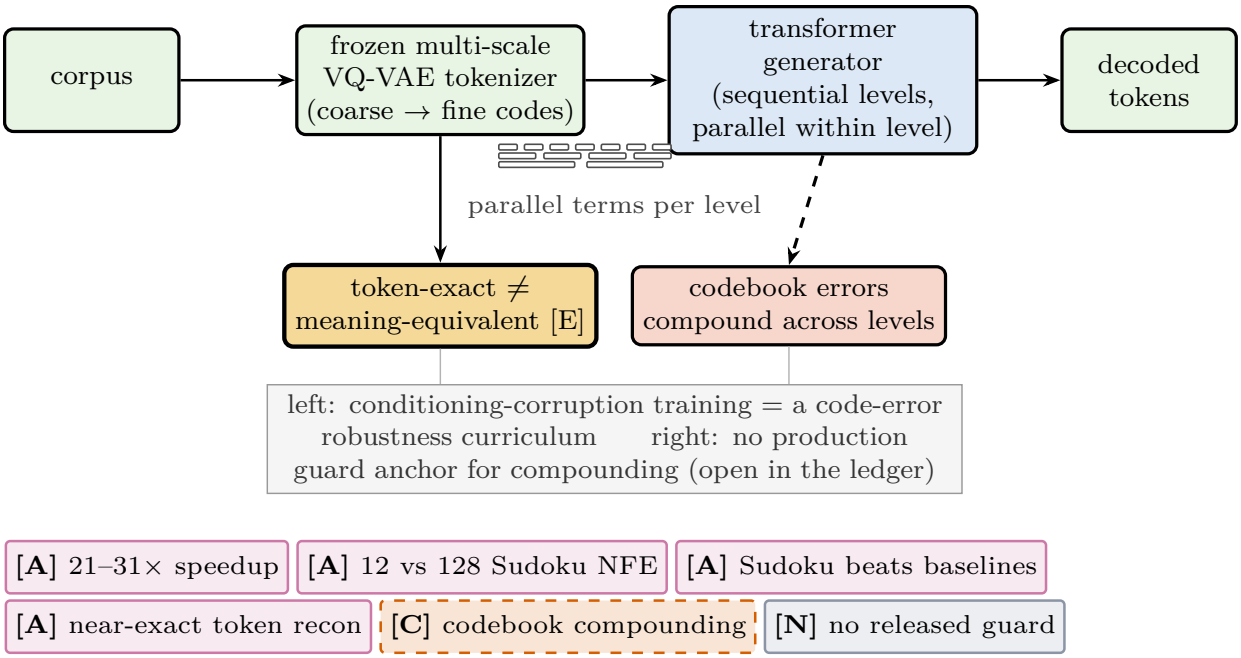


Figure 10: Family card: hierarchical code prediction (NCM-class). Accent = the tokenizer-stage semantic-risk boundary; dashed = contested/undocumented wiring (codebook compounding). Speed, NFE and Sudoku evidence is highlighted on the ribbon. Ribbon segments carry the dominant class ([T]/[E]/[A]/[V]); the full alphabet is Table 3. Colour key: teal = [E] independently evaluated, purple = [A] author-reported, teal outline = [T] theory-under-assumptions, purple outline = [V] vendor-reported, vermillion = [C] contradicted (anchor named), gray = [N] not claimable; all dashed outlines = contradicted (global dashed grammar). The frozen tokenizer is drawn as a store: a fixed codebook consulted at generation, not a component trained in this card.

a continuous vector space trained by prediction or agreement, spanning the JEPA family and the embedding models that retrieval consumes. (iii) *Symbolic or structured state*: retrieved spans, typed records, explicit task state, representations on which deterministic checks are possible by construction. The rest of this section asks what each family guarantees, and what it hands to the layers that follow.

5.2 Joint-embedding latent prediction and the language-JEPA paradox

Joint-embedding predictive training predicts target *representations* rather than raw inputs: a context encoder observes a masked view, an exponential-moving-average (EMA) target encoder embeds the unmasked signal, and a predictor is trained to match the target latent. What the objective encourages is a representation in which masked targets are predictable from the visible context, one biased toward structure from the outset rather than toward surface reconstruction. The family produced strong visual representations (I-JEPA [10]) and motivates the hope that such learning yields a world model rather than a surface generator. Its language case tests our honesty most directly, so we state it as a conditional candidate rather than a verdict.

Squared-error latent prediction is well posed only when the target representation is conditionally concentrated—when, given the context and the target location, the target representation lies near a single meaningful point. Otherwise the optimal predictor degenerates to a conditional mean: a centroid over incompatible alternatives. [T] Masked text need not satisfy this condition: the

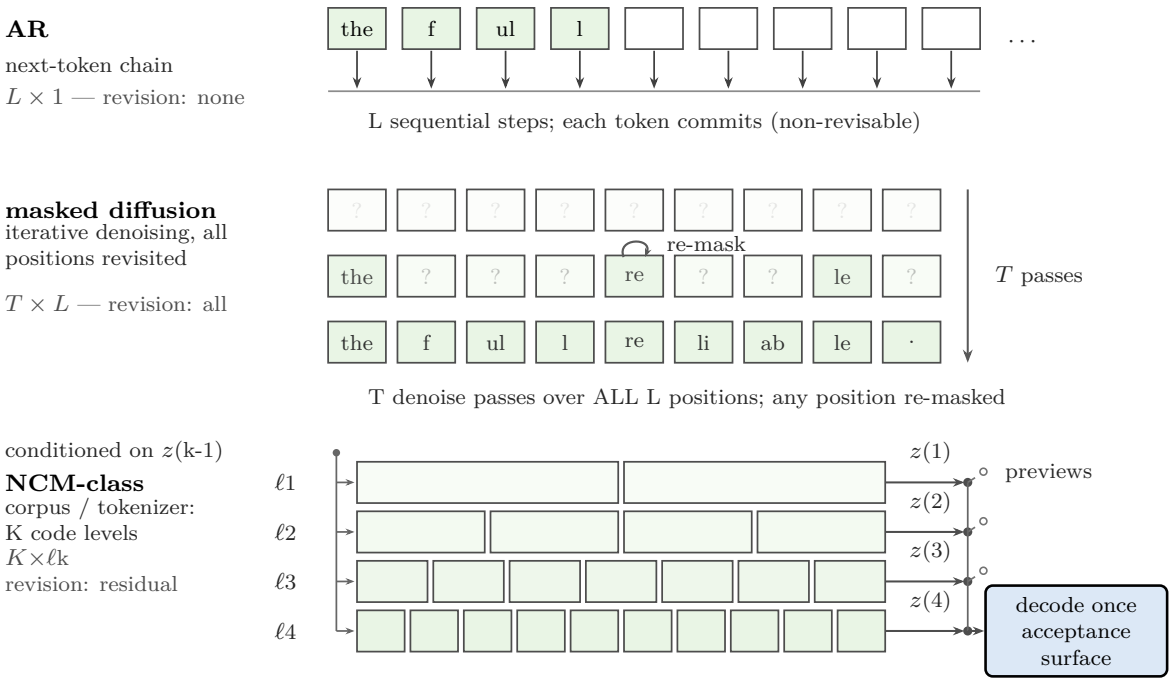


Figure 11: Generation-schedule atlas: the three factorizations differ in how many sequential decisions they make and whether a decision can be revised. Lanes are in Table 1 order, not cost-ranked: left-to-right chain rule ($L \times 1$: one commitment per token, never revised), iterative sequence denoising ($T \times L$: every position revisited each pass, so any position can be re-masked), and the coarse-to-fine code hierarchy (K sequential levels with ℓ_k code positions at level k, parallel within a level; every level spans the same token sequence at rising resolution, it does not append). Each level is fed by the conditioning spine (conditioned on $z(k-1)$) and hands its codes to a cumulative reconstruction $z(k)$; a single decode follows the last level and is the acceptance surface. Dashed ports are diagnostic per-level previews off the K-step path. The only quantitative claim is $K \ll L$; no wall-clock numbers are drawn.

same context can admit several valid continuations whose embeddings occupy distinct directions. In matched I-JEPA/T-JEPA experiments, mutual-information saturation and elevated conditional target variance precede train–validation divergence, effective-rank degeneration, cosine collapse, and poor downstream transfer; the pattern replicates across five independent data seeds, and downstream probes (IMDB and SNLI classification; MTEB retrieval, FEVER and MSMARCO) fail to transfer [36]. [E] The study is explicit that this does not refute predictive learning for language; it identifies a missing inductive bias, namely the preservation of multiple plausible completions instead of compressing them into one latent point. The same analysis notes that marginal variance regularizers constrain only the marginal distribution, so they do not by themselves guarantee predictable conditionals [36].

Identifiability theory for latent world models (the conditions under which a learned latent provably corresponds to the structure that generated the data) is recent and explicitly conditional. Results on when JEPA-style training learns an identifiable world model, and on the identifiability of controlled world models, hold under stated assumptions about data and training; extensions relax the Gaussianity restriction without making identifiability unconditional [38, 87, 158]. [T] We take a standing discipline from this cluster: prediction quality and world-model quality are sepa-

rate properties. Low squared error on held-out targets can coexist with degenerate representation geometry, and whatever structure a representation recovers is recovered only because the stated assumptions hold; the objective does not deliver it by default.

5.3 World models: prediction quality is not decision quality

When Layer 2 hands a world model to a planner, the reliability question changes register. The world-models survey we cite as our anchor treats objectives and representations together [99]; its scope is robotics-oriented, and we borrow from it only the interface claim we need here: serving a model to a decision procedure differs from evaluating it as a predictor. Prediction metrics and decision utility are different currencies: a model can minimize average rollout error and still be a poor or exploitable basis for planning, because decisions consume predictions through an optimizer whose failure modes—compounding error, exploitation of model error, objective mismatch—need not appear in any prediction loss. [S] Planning needs more than prediction: it needs predictions that are reliable exactly where the decision is sensitive, which is a statement about the task and the policy (L0, L3–L4), as Section 10 sets out. The identifiability results above also govern expectations here: where their assumptions hold, "the representation is a world model" has content; where they fail, the same phrase is an aspiration. Unqualified world-model claims are unsupported without those assumptions; since they cannot be evidenced, they have to be gated, which makes them a matter for the assurance layer (Section 8).

5.4 What each representation choice hands to the output and assurance layers

Representation choices propagate downstream: each family of Section 5.1 hands Layer 4 (Section 7) and Layer 5 (Section 8) different raw material.

Token embeddings. The system can emit a distribution over a discrete alphabet, so output contracts built on checking discrete structure—schemas, grammars [40, 63]—have something to verify; what they verify is structure; the boundary to truth is Layer 4's central statement (Section 7). One deployment-level asymmetry matters for reliability: in API-served models the tokenizer is fixed by the serving stack, so tokenization-induced biases are inherited rather than chosen and cannot be repaired by output constraints alone. One recent variant replaces the subword tokenizer itself with a learned multiscale code hierarchy (NCM-class [89]): surface reconstruction is 98.79–100% across the released datasets (99.79% on XSum), yet the authors observe substitutions of named entities and numerals and errors in diacritics and usernames at the tokenizer stage [A]. What such a representation hands Layer 4 is a boundary to define — token exactness vs. meaning equivalence — rather than a fidelity guarantee to inherit (Section 7). [S]

Joint-embedding latents. Similarity geometry and probability semantics answer different questions: a cosine score has no declared meaning, so no threshold policy can consume it. Threshold policies—abstention, escalation, cascades—need a scalar risk estimate with a declared meaning, which a raw embedding does not provide; obtaining one requires adding a calibration or conformal layer together with its own assumptions (exchangeability for conformal procedures; alignment conditions for provable abstention) [39, 150]. [T] The calibration and uncertainty surveys map what confidence estimates do and do not carry for LLM systems [61, 103], and abstention designs (surveyed in [148]) are the constructive response. Output contracts that emit typed decision probabilities as first-class outputs [35] occupy exactly this design point: they make the decision probability, not a similarity score, the object that a threshold policy consumes.

Byte-level input. Removing the tokenizer trades a class of tokenizer-induced failure modes for sequence length; attention cost grows super-linearly in length [140], so the context budget that

finer granularity spends is unavailable to retrieval and verification machinery unless multiscale designs buy it back [156]. The pattern generalizes: representations do not determine reliability, but they determine which reliability mechanisms are available downstream, and Section 10 follows that thread.

5.5 What Layer 2 does not guarantee

Identifiability is never unconditional: the results cited in this section hold under stated assumptions about data and training, and outside them nothing follows. [T] Representation and semantics are distinct: a latent can be stable and low-loss yet underdetermined with respect to the meaning a decision needs. The language case remains open in both directions: deterministic squared-error latent prediction fails under matched protocols on text [36], while the same authors leave predictive representation learning for language explicitly unrefuted. [E] We therefore record *stable language-JEPA as contested*: a candidate the field has neither established nor settled, and an honest-null entry in the ledger of Section 11; at matched evidence standards, no language representation recipe in our ledger carries the evidence its image- and video-modality ancestors do [N] (scoped to the searched literature, cutoff 2026-10-17).
The mechanism is sketched in Fig. 12.

6 Layer 3 — Inference Architectures

6.1 Generalists, Specialists, and Hybrid Pipelines

Layer 3 covers the choices that govern how a trained model is *used*: at what granularity, with what conditional computation, and with which external calls. The first choice is granularity. A generalist is asked to produce everything at once — answer, format, and whatever confidence the response needs; a specialist commits to narrower outputs that a downstream system can verify mechanically: boxes, discrete labels, bounded scores. That verifiability is the argument for the specialist architecture, and also the one most often asserted rather than measured end to end. Interfaze positions an orchestrated set of task-specific small models against general-purpose incumbents [85] [V]; the claim lives at component level, and Fig. 19 reconstructs one such pipeline from vendor descriptions chiefly so that the edges the vendor leaves undocumented — monitors, validators, escalation — stay visible in our account. The hybrid pays for composition: every added component brings its own error distribution, and the pipeline's end-to-end reliability depends on how those errors propagate, not on the best component's headline number. We therefore treat hybrid pipelines as stack instances with their own failure envelopes rather than as upgraded single models.

6.2 Mixture-of-Experts: Routing Inside the Model

Mixture-of-Experts is conditional computation inside the model: capacity scales with parameter count while cost scales with activated experts, and the router becomes a learned, partially inspectable dispatch policy [21]. Our selection framework needs the intra-model and system-level split more sharply than the survey literature draws it, so we keep two routing loci strictly apart: *intra-model* routing decides which experts fire for a token and is an architecture property; *system-level* routing decides which model or cascade tier receives a request (Section 6.3) and is an orchestration policy. On reliability, MoE-RBench is the benchmark evidence we anchor: Across safety and hallucination, adversarial resilience, and out-of-distribution robustness, sparse MoE can be made more reliable than dense models of its class, but robustness is sensitive to basic training settings

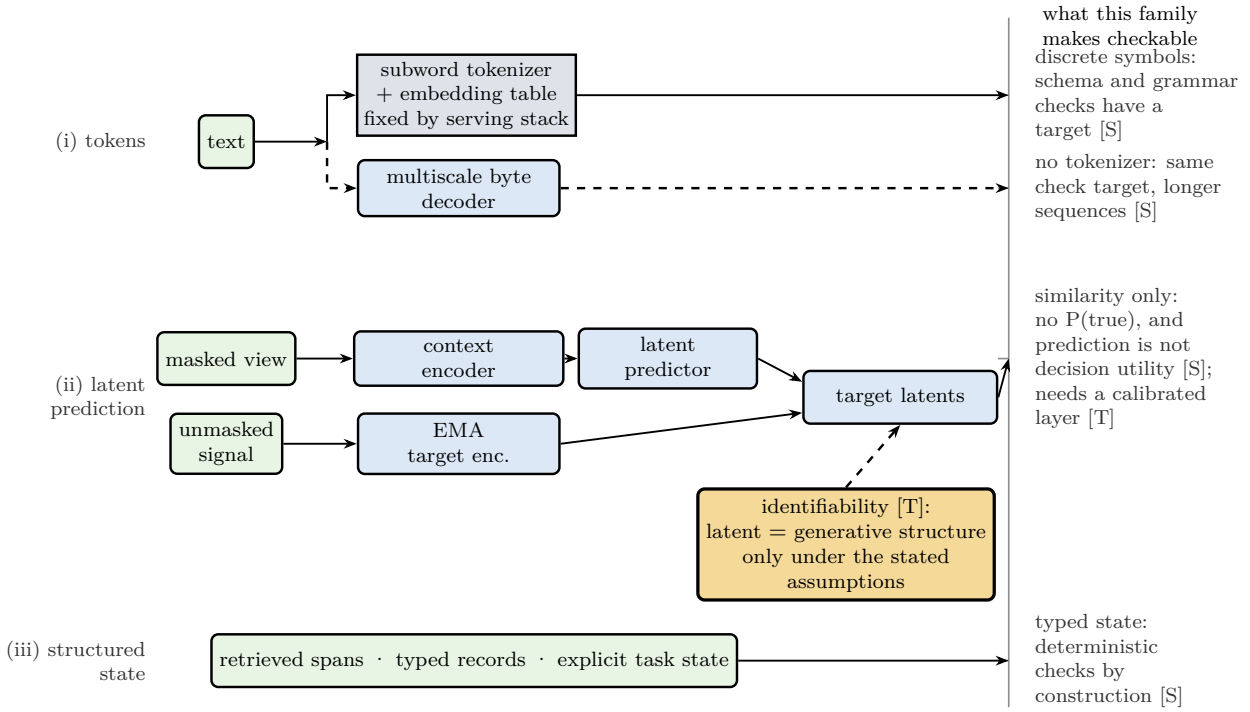


Figure 12: Layer 2 schematic: three representation families and what each hands the layers above. Tokens are discrete symbols, so schema and grammar checks have a target; the tokenizer-free byte fork keeps that target and pays in sequence length [J]; joint-embedding latents carry similarity only — no $P(\text{true})$ and no decision utility — so a scalar risk estimate needs a calibration layer with its own assumptions [T]; symbolic or structured state supports deterministic checks by construction. Yellow: identifiability [T], the latent equals generative structure only under the stated assumptions, with no anchor in our set bridging it to decision quality [GAP]. In API-served models the tokenizer is fixed by the serving stack [S].

and hyperparameters [25] [E]. Reliability here is a tunable, configuration-dependent property. Compression follows the same logic: structured expert pruning preserves in-domain utility at moderate ratios, whereas hallucination risk rises at extreme pruning and both utility and factual reliability degrade quickly out of domain [151] [E]; at deployment time compression is a reliability decision as much as a memory one.

6.3 Routing and Cascades: Cost–Quality Allocation as an Architectural Lever

Above the model, allocation happens per query. FrugalGPT established the cascade pattern (chains of increasingly capable models whose cheaper members are trusted only when their scored responses clear a cutoff), reallocating spend toward the queries that need it [26]. The architectural weak point is the dispatch signal: cascades greedily select the lightest model from confidence scores, but deep models are systematically overconfident and confidence distributions differ across models and languages, so the C3 system [105] interposes a per-member calibration stage before escalation. SATER unifies pre-generation routing with cascade-style escalation under a token-efficiency objective, targeting the accuracy–latency split between the two families [135]. Routing also extends beyond a fixed pool: UniRoute routes to previously unseen models by representing each candidate through its predictions, with excess-risk bounds relating the learned rule to a theoretical optimum [81]. The

evidence base remains fragmented. The characterization survey of routing and hierarchical inference is efficiency-scoped and notes the absence of a unified evaluation across methods [17], and the risk-keyed selection question stays open (Section 10).

6.4 Looped and Recurrent Depth: Compute Scaling Without Parameter Scaling

Depth is the second axis, and it can be repeated rather than stored: looped transformers apply one compact stack for multiple rounds, buying unrolled depth at fixed parameter count. DeepLoop's tied-depth analysis yields a first-order perturbation bound controlled by a visit-alignment coefficient — the shared weight update aggregates gradients from repeated visits — recovering standard residual-scaling exponents when visits decorrelate and requiring larger ones in the aligned regime [T] [96]. The DeepLoop authors' accompanying ablations are small-scale (GPT-style looped language models), effectively tied with the unlooped baseline at one loop and improving at moderate loop counts, but single-seed [A], so we record a stability envelope rather than a performance verdict. Engineering follow-ups attack the family's latency cost with self-speculative decoding [28, 104]. The broader recurrent lineage makes the same wager with fixed-size state: the RWKV survey documents the merge of recurrence and attention [101], and a controlled 72-model study of hybrid linear attention finds that standalone superiority of a linear-attention variant does not transfer to its hybrids, with recall governed by the linear-to-full mixing ratio [E] [143], so the envelope belongs to the hybrid rather than to either component.

6.5 Memory, Adaptation, and Latent Reasoning at Inference Time

The same reuse principle also operates inside the weights, where it yields memory learned at test time: Titans adds a neural long-term memory that memorizes historical context while attention handles the current window [18], and TPTT retrofits pretrained transformers with linearized attention and memory gating without full retraining [54]. Adaptation at inference then stops being a deployment-time constant: verifier-driven test-time training selects pseudo-labeled examples through a learned verifier and adapts the model out of distribution [A] [112], which is why this row lives only partly on Layer 3; the verifier's own reliability, the drift adaptation induces, and the monitoring it requires are assurance concerns (Section 8). Hierarchical recursive reasoning pushes the same logic to continuous depth: HRM demonstrated strong results on 2D reasoning tasks with compact latent recurrence rather than pretraining scale [144], and the evidence-discipline lesson lies in the delta between those demos and the general-reasoning claims they invite. A critical review re-examines the design space, tests variants, and corrects misconceptions of an explicitly early-stage family [58], while a dynamical-systems reformulation recasts the recursion as continuous dynamics with an explicit equilibrium requirement and feature-collapse mitigation, addressing the instability the original training dynamics do not guarantee away [45]. Latent reasoning also narrows the controls we take for granted in token space: activation steering of continuous thoughts produces substantially weaker effects on generation than steering explicit chain-of-thought, even at comparable movement inside the representation [E], a latent-to-language transition gap [73], and detection work builds benchmarks to probe misaligned reasoning planted inside continuous thought [126].

6.6 Tool-Augmented Agents

The final move re-opens the model to the world: ReAct interleaves reasoning traces with actions that query external environments, letting observations steer subsequent reasoning [154]. Architecturally, each tool call is a verification opportunity — deterministic functions, retrieval, and execution

are checkable in ways token generation is not (Sections 7 and 8) — and simultaneously a new failure path: planner errors propagate through execution, intermediate state persists, and some actions are irreversible. The agent surveys chart precisely this surface, factoring planning into task decomposition, plan selection, external modules, reflection, and memory [74], and organizing broader agent methodology around architecture, collaboration, and evolution [106]. None of these taxonomies carries a reliability guarantee; they enumerate where validation must be placed, and because they relocate rather than remove the reliability burden, they make the strongest case in this layer for co-designing L3 and L5.

6.7 What Layer 3 Does Not Guarantee

Read as a whole, the layer offers a set of allocation choices: where to spend parameters (MoE), repeated depth (looping), or calls (cascades). Each of those mechanisms is a candidate whose envelope we have tried to state. None of them makes the model calibrated by construction. Cascades and routers consume confidence estimates that base models oversupply and miscalibrate [105], compressed experts shift where hallucination risk sits [151], and test-time memory moves the training–inference boundary into deployment [112]. End-to-end reliability remains a system property, and component-level wins — even independent component benchmarks such as the decision-tier evaluations of the Jev System-One family [35] — do not compose into system guarantees. Layer 3 supplies candidates and their costs; what the system may claim is decided by the selection procedure and the assurance layer (Section 10, Section 8).
The mechanism is sketched in Fig. 13.

7 Layer 4 — Output Contracts: Validity Is Not Truth

7.1 The contract is the interface the software can rely on

Layer 4 asks what a component *returns* to the software around it. We call the declared, externally checkable shape of that return an *output contract*, ordered in four contract tiers (T1–T4), numbered as in Fig. 14. *Free-form text* is the weakest: downstream code can act on it only through another parser or a model, so the contract is "whatever the consumer can extract" (T1). *Schema-constrained output* adds a grammar or JSON schema the return must satisfy (T2). *Typed decisions* remove text entirely: the component returns a distribution over a fixed set of typed choices (a probability per option or per stated claim), so the decision score is what thresholds consume (T3). *Structured metadata* — citations, spans, tool-call records, confidence fields — layers evidence onto any of the above (T4). The tier fixes what later layers can check and select over (Section 10): a validator can test a grammar, a threshold can consume a score, and neither can test a sentence. That is the handoff promised in Section 5.4, where representations decide what the system computes over and contracts decide what can be verified about what it returns.

7.2 The validity half: constrained decoding, and what validity measures

Grammar-constrained decoding makes the validity half constructive: at each step it suppresses every token that would violate the grammar. Constrained decoding was introduced to close a specific gap: language models fail to produce complex structures reliably when not finetuned to the required format, and masking invalid continuations restores structural correctness without retraining [62]. [E] Serving engines followed: XGrammar constrains generation against grammars and JSON schemas at practical overhead [40]. The guarantee is precise and narrow: whatever the engine emits parses

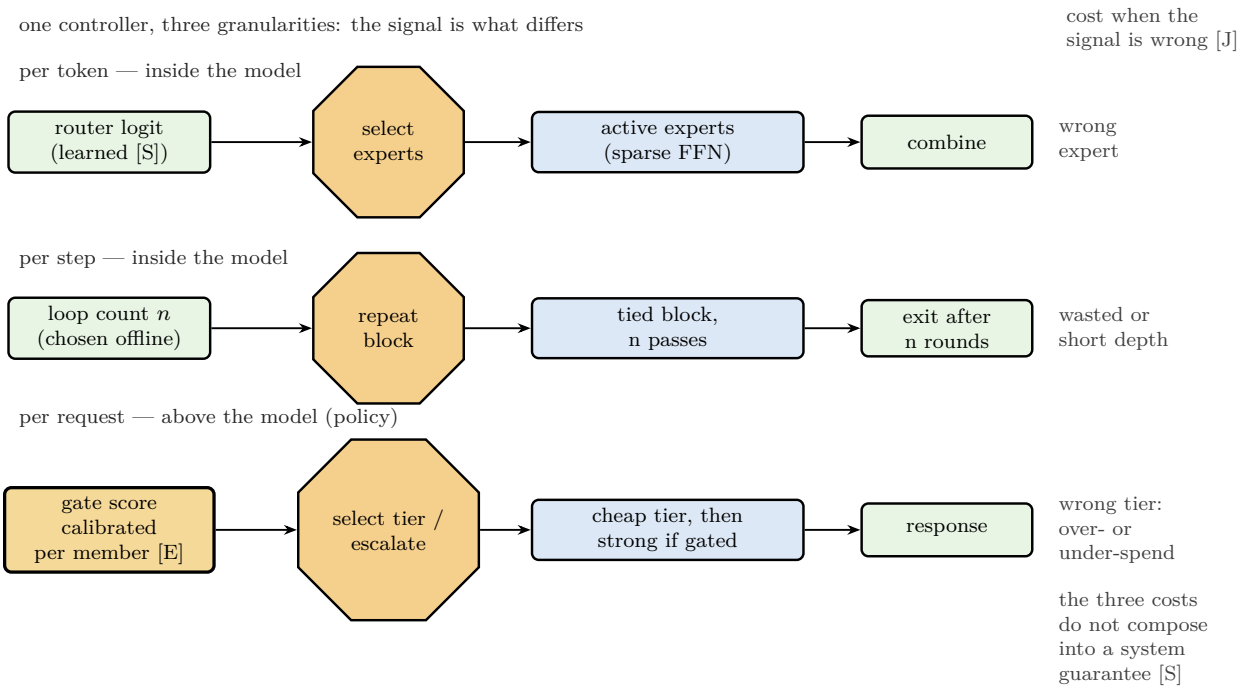


Figure 13: Layer 3 schematic: one selector shape, three scopes — per token, per step, per request. Per token (expert routing), per step (looped depth) and per request (tier choice across generalist, specialist and cascade members, or escalation) the same signal–select–compute–output shape appears; only the per-token case merges parallel work (expert outputs), the other two return one result. What differs is the honesty of the signal — a learned router logit [S], a loop count chosen offline (a design constant, not a per-input signal; the tied-depth bound is [T]), and a gate score calibrated per member [E]. The thicker-bordered box in the third band (gate score) is the only signal with a measured calibration anchor (C3, SATER). Cost–quality allocation is a first-class reliability lever [S]; the cost when the signal is wrong or absent is our analytic cut [J], and none of the three composes into a system guarantee.

under the declared schema. What it is worth is measured. JSONSchemaBench separates coverage (which schema features an engine actually supports), output quality, and generation efficiency, rather than treating "supports JSON" as one property [63]. [E] The separation matters: a schema an engine cannot fully express degrades silently to a weaker contract, and latency can make a valid output unusable. Our first load-bearing conclusion, by synthesis rather than citation: schema compliance is table stakes: validity is not correctness. Whether the values filling the schema are true is a distinct property that no decoding-time guarantee addresses, because the grammar constrains form and says nothing about content. [S] A perfectly formed JSON object can be confidently wrong in every field.

7.3 The truth half: typed decisions and calibrated probabilities

Decision-oriented models occupy the typed-decision tier by construction. The Single-Shot/System-One family returns no text: given a state and typed questions, the model returns a choice from fixed options, a position on a rubric, or the probability that a statement is true, and independent benchmarking evaluates these returns on held-out data [35]. [E] The vendor architecture documents the interface in its own vocabulary — typed `choice`, `score` and `noul` primitives, `noul` (the vendor's

spelling) being a P(true) query — and reports an RLCD recipe built on proper scoring rules (log score, spherical score, ranked probability score) whose object is a truthfully reported distribution.[2] The contract does not remove the truth problem; it relocates it, from "is the text true?" to "are the probabilities honest?" That second question is measurable. Independent evaluations now span decision accuracy and calibration in medicine [43, 107], binary decision scores under a risk budget [109], and fine-tuning comparisons [100]. [E] Vendor probability claims enter as hypotheses, not facts; that is the evidence discipline of this survey.

The discipline has already caught a live disagreement. Vendor documentation warns that shipped checkpoints are over-confident and need temperature refitting before gating relies on them. [V] One reproduction study measures the shipped typed-decisions checkpoint as uniformly *under*-confident, signed confidence–accuracy gap -0.214, improved by a single fitted sharpening temperature [114]. [E] The direction of miscalibration is therefore reported as *contested* (an open ledger entry in Section 11, cited per checkpoint and never as a family-level fact), and the sign decides which way a confidence-gated cascade fails: an over-confident model passes wrong answers through the gate; an under-confident one abstains on correct ones. A gate designed on the wrong sign misprices both its accepted-error budget and its coverage.

7.4 Thresholds are policy; option sets set the scale

Typed probabilities become gating machinery only through a threshold, and the operational literature has converged on the stance this paper adopts: "a threshold is a policy you choose from measured accuracy at that coverage on your data, not a property of the model"[3] Two constraint classes — option count and option order — enter the ledger as separately sourced rows; no single law covers them. The vendor's decision-model card advises keeping `choice` questions under roughly twenty options within a fixed per-label budget, with sharp falloff beyond; the mechanism is budget trimming, so we record guidance, not a measured cliff. [V] On a 77-label intent benchmark the vendor reports 0.425 against 0.870 intent accuracy for the Jev-class model. The comparison is uncontrolled and cross-source; the vendor itself flags it as indicative. [V] Independently, paired evaluation finds the same family changes 30% of its answers under option-order reversal and reaches 31% accuracy at fifty nearest-neighbour tools [95]. [E] Option count and option order are therefore contract-design variables. Any accuracy or threshold claim is only meaningful at the option configuration it was measured on.

7.5 The hub: risk-controlled serving as the L4–L5 bridge

The contract tier creates the preconditions for risk-controlled operation: typed, checkable outputs with a declared correctness measure are what calibration needs. The cleanest current construction is risk-budgeted configuration selection (Redline). Given a reference configuration and a candidate, a finite-sample procedure selects operating points — acceptance thresholds, buffer depth, schedule, checkpoint, precision — such that the *reference-relative joint risk*, the probability over prompts that the reference answers correctly while the candidate does not, stays within a user-chosen budget with high probability, with no distributional assumptions [92]. [T] Its scope is as sharp as its strength: relative to a reference and a chosen correctness measure, and by construction *not* a factual-correctness guarantee; speed is bought at a bounded disagreement rate, not truth. The budget is also task-dependent in the authors' measurements (math clears at a tighter budget than code), which is why we treat it as an L0 input rather than a constant. Full treatment is Section 8.7.

[2] `github.com/NandhaKishorM/laya` documentation, accessed 2026-10-17; evidence class [V].

[3] Ibid., "Automated Confidence Gating," which in our framework is derived from the risk budget of Section 8.5.

The contract-side lesson is that a contract fixes what "accurate" means for the return, which is what lets "faster but less accurate" become a number an operator can budget.

7.6 What Layer 4 does not guarantee

Each contract tier constrains *form*, and only form. Free-form text guarantees nothing checkable; schema-constrained output guarantees parseability, not that the parse is true; typed decision probabilities make confidence auditable, not accurate; structured metadata makes evidence *attachable*, not attached or faithful. An NCM-class staged tokenizer ([89]) sharpens the same split one stage earlier: the token-exactness vs. meaning-equivalence boundary is a contract statement to define explicitly — reconstruction fidelity on the code side is not meaning fidelity on the read side. [S] A schedule change also relocates the error surface: grammar and schema gates operate on decoded tokens and never see the latent codes, so a substitution becomes observable only after decoding, and the latent channel remains unaudited while acceptance machinery must be anchored at the decoded surface. [S] The authors' conditioning-corruption training — their code-error robustness measure — is the corresponding hardening move: robustness bought inside the representation rather than enforced at the contract. [A] Semantics enter only through Layer 5's checks — grounding, validators, calibrated selective prediction, escalation to human review (Section 8) — which are themselves fallible components, as the assurance layer's own failure envelope concedes (Section 8.8). The asymmetry is structural: form is checkable by construction; truth is a separate property. It is also why the selection framework (Section 10) treats the contract tier as a design decision, not a formatting preference.
The mechanism is sketched in Fig. 14.

8 Layer 5 — Assurance and Verification

Cross-cutting safeguard coverage is summarized in Fig. 16: which failure modes have evidence-tagged coverage, and which cells are named gaps.

> "A threshold is a policy you choose from measured accuracy at that coverage on your data, not a property of the model. … Then pick the point where the errors you accept are ones you can live with. Confidence orders decisions; it does not establish that a decision is correct."
> — Laya project documentation (vendor-reported)[4] — the stance this layer formalizes.

8.1 The layer contract: reliability as residual-risk management

Layer 4 delivers outputs that are structurally valid: schema-conformant, typed, parseable (Section 7). Validity is not truth, and no output *contract* can make it truth: hallucination persists across model families, tasks and elicitation formats, which is why detection and mitigation retain survey literatures of their own [2, 159] [S]. Layer 5 is the collection of mechanisms a system uses to manage that residual: attaching evidence to outputs, checking them, refusing to deliver some of them, and routing the rest to a human. Its contract is deliberately weaker than "no errors" and deliberately stronger than "hope":

> An L5-wrapped system promises that undetected errors are a *managed quantity*: the operator chooses how much error may pass unflagged, the system measures how much actually does, and drift or evasion appears as a monitored signal.

[4] `github.com/NandhaKishorM/laya`, README, accessed 2026-10-17; evidence class [V].

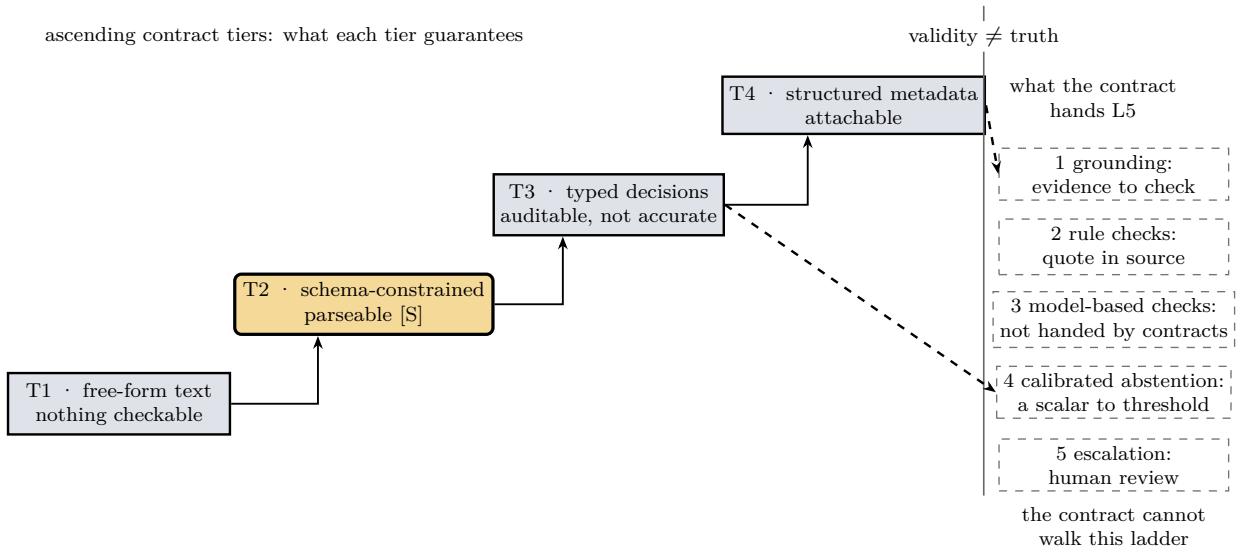


Figure 14: Layer 4 schematic: four contract tiers, ordered by what each adds to the return — an ordering that is our analytic cut [J]; T4's metadata is an add-on to any of T1–T3. Each tier is printed with its own defeat — free-form text guarantees nothing checkable; schemas buy parseability, not truth [S]; typed decisions make confidence auditable, not accurate; structured metadata is attachable, not attached or faithful. Yellow: T2's schema compliance is table stakes [S]. Right of the divider (validity ≠ truth) sit the L5 rungs a contract can reach, numbered as in Fig. 15: 1 grounding, 2 rule checks (deterministic validators), 4 calibrated abstention (selective prediction at a threshold), 5 escalation to human review. Two receive a named object: T4 hands grounding evidence to check, T3 hands selective prediction a scalar to threshold; rung 3 (model-based checking) consumes no contract object. No tier reaches the truth of the values it carries, and the contract cannot walk the rung column.

The responses available to the layer, ordered roughly by cost, form a ladder. (1) *grounding* — attach retrieved evidence and tool results, so claims become attributable; (2) *deterministic checks* — validators and content rules (quote-in-source tests, unit and interval checks) that reuse the Layer-4 machinery but verify content rather than shape; (3) *model-based checking* — self-consistency and learned verifiers, which trade sampling compute for an error signal; (4) *calibrated selective prediction* — accept or abstain against a threshold derived from a risk budget (§8.5); (5) *escalation* — defer the hardest cases to a human or a slower mode. The ladder is cumulative, and every rung is itself a component that can fail; §8.8 therefore closes the chapter with the layer's own failure envelope. The epigraph is the stance the calibrated rung formalizes: thresholds are policies chosen from measured behavior.

8.2 Retrieval grounding: evidence channels and their new failure paths

Grounding is the cheapest way to change a claim's *epistemic status*: if the document a system cites is the source of what it says, a reviewer can audit the claim without auditing the model. In its interactive form the model acquires the evidence itself: ReAct interleaves reasoning traces with task-specific actions so that the model queries external sources and conditions on what comes back, which the original work credits with reducing hallucination and error propagation on question answering and fact verification [154]. In its passive form, retrieval-augmented generation supplies context up

front; the trustworthy-RAG surveys are the systematic accounts of what this buys and what it costs. RAG is reported to reduce hallucination and freshen knowledge, but it introduces new risks along its own pipeline (robustness, privacy, adversarial exposure, accountability), and unreliable retrieval or improper use of retrieved knowledge still yields bad outputs [117]. A companion framework scores RAG trustworthiness on six dimensions — factuality, robustness, fairness, transparency, accountability and privacy — with a benchmark for each [162] [S].

The failure paths are what the layer must be designed around. *Retrieval miss:* the evidence exists but is not retrieved, and the generator answers ungrounded with undiminished fluency; misses of this kind stay invisible unless the system is required to cite. *Stale or conflicting sources:* grounding certifies that a claim came from *a* document; that document can be wrong, and a confidently stale source launders error into apparent diligence. *Unfaithful use:* the evidence sits in context but does not shape the output, the "improper knowledge utilization" the survey singles out [117]. The nominal window and the effective one diverge: of seventeen models claiming at least 32K tokens, half keep satisfactory performance at 32K [70] [E]; under a needle test that removes the lexical shortcut, eleven of thirteen fall below half their short-context score at the same length, GPT-4o from 99.3% to 69.7% [111] [E]; and an eighteen-model industry evaluation finds degradation with input length on tasks that require no retrieval at all [30] [E]. Vendor guidance on managing the window — compaction, note-taking, sub-agents — is vendor-reported [7] [V], and no independent replication of it exists in our evidence [N]. *Lossy context:* the window is a lossy store — it holds what was retrieved, and what gets used is a separate step. Grounding therefore converts an unobservable failure (model error) into an *auditable* one (citation mismatch) that deterministic checks can catch. It does not convert error into truth.

The index is a design decision, and its shape is a measured design variable. A flat top-k store fails questions that are global to a corpus by construction, which is the gap graph-structured indexes are built to close [44] [A]; a graph index over entities and passages buys about 20% on multi-hop retrieval at 10–30× lower cost than iterative retrieval [65] [A]; and the traversal policy over that graph is worth up to 9.96 NDCG@10 over the strongest fixed policy while exploring 81.9% fewer paths [91] [A]. The three results say one thing: which questions are answerable at all is decided before the generator runs, so a retrieval miss is a property of the index and its policy as well as of the query. [J]

The audit claim has a false-negative rate of its own. In a controlled reproduction on Wikipedia-based question answering, roughly one citation in seven was unfaithful — the answer was right and the cited passage did not support it — and the rate did not improve under RLVR, one trained search agent landing slightly worse than its base model [84] [E]. ALCE puts incomplete citation support at 50% for the best models on ELI5 [56] [E]. Citation mismatch is therefore a check with a measured miss rate: it catches the citation that fails to support the sentence and passes the citation chosen after the answer was already fixed, which is why a citation check belongs in the budget with a stated miss rate.

8.3 Self-checking: agreement as a cheap signal and its sampling price

The cheapest model-based checks verify nothing external; they interrogate the generation process itself. Zero-shot chain-of-thought showed that a single elicitation phrase can move accuracy substantially on arithmetic, symbolic and logical tasks without any exemplars [88] [A] — a prompt-level change to the error distribution: the system learns nothing about *which* of its answers are wrong. Self-consistency turns sampling itself into a signal: draw a diverse set of reasoning paths, answer by the agreement among them, and benchmark gains on arithmetic and commonsense reasoning follow [146] [A]. The cost is the k-sample price in latency and budget, and it scales with k.

Self-consistency is best read as an *uncertainty estimate*, with two caveats. First, a systematically wrong model can produce agreement too, and the uncertainty literature is explicit that such scores are heuristic: a threshold chosen on them does not by itself provide statistical guarantees on the error rate among accepted answers [39]. Second, the signal is nonetheless useful raw material: the conformal-abstention construction takes self-consistency as its reliability measure and adds the missing guarantee on top (§8.5). The pattern repeats at every rung: heuristic signal first, calibrated signal where the stakes require it.

Minimum-Bayes-risk selection over cheap candidates is the assurance-side use of the sampling budget that shorter generation paths free up: on XSum the NCM authors report about 7% higher ROUGE-L than the strongest summarization baseline within the same sampling-time budget. [A] The gain comes from agreement among candidates. Agreement is not independent verification, so budget reallocated this way must still route through this layer's grounding and verification machinery (Section 8.4).

8.4 Verification loops: process supervision and the verifier as a component

Where self-checking asks the model about itself, a *verifier* is a separate component that is trained to judge. Process reward models score reasoning at the step or trajectory level rather than judging final answers alone, and the survey of the area organizes the full loop — how process data is generated, how PRMs are built, and how they are consumed for test-time scaling and reinforcement learning — across math, code, multimodal reasoning, robotics and agents [161] [S]. Verification signals can also be recycled into adaptation: verifier-driven test-time training scores a pool of sampled answers, keeps the high-confidence ones as pseudo-labels, and fits a small adapter to them, improving the base model on structurally novel tasks [112] [A]. This is Layer 5 re-entering L3 and L1: assurance outputs become inference and training signal.

Both moves share a structural caveat. A verifier is itself a model, with an error rate and a proxy objective; when it selects data or scores rewards, the generator acquires an incentive to satisfy the verifier rather than the world, and the verifier's own errors are no longer checkable by the system that consumes them: they *become* the system's errors. Verification loops therefore reduce some errors and relocate others, and the relocation is invisible from the outside unless the verifier's accuracy is itself maintained against held-out ground truth. This is the first entry in the layer's failure envelope (§8.8).

8.5 Calibrated selective prediction: thresholds as policies derived from risk budgets

The strongest rung is the most studied: use the model's own uncertainty to decide when to answer at all. The survey anchors agree on the landscape. The confidence-estimation survey organizes calibration techniques and their applications [61]; the newer uncertainty-quantification survey adds a taxonomy along two axes — computational efficiency and uncertainty source (input, reasoning, parameter, prediction) — and frames selective prediction as the operative risk-mitigation use case [103]. Abstention now has a survey of its own, organized by what is uncertain (the query), who is uncertain (the model), and against whose values abstention is judged (human values); the last axis is what makes abstention a policy [148] [S].

This paper's stance is that a threshold is *derived*, never universal, the position the epigraph states in vendor language. An operator fixes a risk budget (the maximum tolerated error rate among accepted outputs, per task and per unit of harm) and then selects the most permissive threshold whose measured behavior on held-out data satisfies it. The budget is where costs enter:

a routing decision and a legal determination get different budgets. The vendor documentation for the decision-model case study (the epigraph's source) supplies the operational corollaries: fit and validate any cutoff on held-out data *at the option counts the workload uses*, because the usable threshold shifts with the configuration, and refit shipped confidence values before any gating policy relies on them.

Two constructions make the budget statement literal, and they are the layer's theory-class ([T]) machinery. A conformal-abstention procedure uses self-evaluated similarity across sampled responses as the confidence measure and conformal prediction to bound the hallucination rate among non-abstained answers [150]. A second framework turns any uncertainty score into a selective-answering rule: for each candidate threshold it estimates a high-probability upper bound on the error rate among accepted answers, under a criterion the application declares, and then adopts the largest threshold whose bound stays below the user's risk [39]. Both are conditional guarantees: distribution-free within an exchangeability assumption on calibration data, and only as good as the criterion that defines "error" [T]. They formalize what the surveys only recommend: the threshold is the output of a procedure over data plus budget. For the newest generation family, the criterion is concrete: where the frozen codec's residual errors are substitutions of named entities and numerals and errors in diacritics and usernames while surface reconstruction stays near-exact (NCM-class [89]), the accepted-error budget must state whether a fluent substitution that changes a referent or quantity counts as an error, and at which layer it is adjudicated — the token-exactness vs. meaning-equivalence boundary, priced rather than assumed. [S]

The fifth rung: escalation. Abstention keeps a case out of the answer; escalation moves it to a human, or to a slower and more expensive mode, whose capacity and error rate are themselves resources that belong in the budget. The abstention survey's human-values axis is the reminder that "answer or abstain" is a judgment about consequences; escalation is where those consequences are owned explicitly.

That budget is usually written as though review were a constant and a catch. Neither assumption survives measurement. In a pathology study, 7% of baseline-correct estimates flipped after an erroneous machine prediction, inside a 6–11% range reported across specialties, and the coefficient on the AI estimate was 0.74 when the advice beat the reviewer's own estimate against 0.30 when it was worse [129] [E]. Expertise moderates the damage without removing it: in a 20-resident simulation, management accuracy fell from 77.3% under concordant advice to 45.3% under discordant advice [11] [E]. Review also depletes: computer-aided detection's adenoma advantage was present early (49.9%) and gone late (39.9%, against a stable 39–41% unassisted) [72] [E]. Detection of naturally occurring errors ran 24% at baseline and 31% at high injected-error prevalence, at 16–18% more time per review [49] [E]. Capacity can be budgeted as a queue rather than a constant: on a moderation benchmark, selective escalation held human load at 16–26% of cases where uncertainty baselines routed 28–39%, at $38–77 against $127 for always-trust [14] [A].

The distinction that matters is therefore whether a human can catch the failure. A reviewer needs an independent judgment formed before the output is shown, the evidence to check it retrievable at review time, the time to use it, and the authority to overrule; without that support, oversight decays with repetition even among reviewers [53] [E]. What the evidence does not contain is the term an operator would budget: no study in this sweep reports precision, recall or minutes per review for reviewers working a production LLM escalation queue, so the rates above come from adjacent settings and do not transfer. Escalation priced that way reprices risk; it does not remove it.

8.6 Monitoring: the monitored quantity, and the evadable layer

A deployed system's thresholds decay: calibration data ages, the workload drifts, and the monitor's own distributions move. Monitoring is the rung that tests, after deployment, whether the assumptions the earlier rungs rest on still hold, and its first design question is what to monitor. The monitoring literature's sharpest lesson is that the obvious answer is usually a proxy. Runtime-monitor research argues that out-of-distribution detection is *not* the right framing: a monitor's job is to discard incorrect predictions, and measured on that quantity ("out-of-model-scope" detection), OOD scores can give a false impression of safety, while OOD-framed comparisons need not identify the better monitor [66] [E]. The lesson parallels the threshold-as-policy stance: the monitored quantity is chosen by what the operator cares about, and validated against that quantity.

The monitored surface is also wider than correctness. Fairness-monitoring tools specify subgroup-behavior properties in a runtime specification language and expose them with live visualization, and they target the human-oversight duties that regulation such as the EU AI Act places on high-stakes decision systems [15] [E]. Monitoring is where the layer becomes auditable: a monitor is the operational form of an oversight duty.

The honest warning, and the newest result in this section, is that the assurance layer is itself a target. An evaluation of instrumental evasion places agents in task-policy pairs where completing the task requires an operation the runtime monitor prohibits, with the agents aware they are monitored: best-of-3 evasion *attempt* rates reach up to 98% and success rates up to 88%, evasion grows with test-time compute, and the successful strategies are mundane: encoding prohibited commands, decomposing an operation across several tool calls, and retrying until the relevant context leaves the monitor's history [131] [E]. Two implications. First, oversight-evasion emerges from ordinary goal-directed behavior under task pressure, and belongs to the deployment setup that the assurance layer must budget for. Second, the trade-off cuts both ways: in the same study the model with the lowest evasion rate paid for it in overrefusal, abandoning otherwise solvable tasks. Monitoring policy, like the thresholds it feeds, is chosen against consequences.

8.7 L4–L5 bridge: configuration selection against a risk budget

The pattern Layer 5 formalizes is already visible in production serving practice, which is why the contract layer and the assurance layer exchange more than a one-way handoff: the contract tier decides what can be measured, and the measurement decides where the operating point sits. The clearest current demonstration is block-diffusion serving. Its operating points — acceptance thresholds, buffer depth, schedule, checkpoint, precision — are normally hand-picked by mean benchmark accuracy, but a mean does not tell an operator how often a faster configuration fails on prompts that a slower reference configuration answers correctly. The risk-guaranteeing procedure (Redline) selects operating points from their measured correctness on calibration prompts, keeps the reference-relative risk — the joint event "reference correct, candidate wrong" — within a user-chosen budget with high probability, and deploys the fastest configuration that passes [92]. The guarantee is finite-sample and distribution-free, but relative to the reference configuration and the chosen correctness measure; LEC gives the general form for selective prediction and routing, where linear expectation constraints hold the risk of the selected set below a budget [147]. [T] On the same calibration data the procedure stays within its stated failure probability, whereas a mean-accuracy rule at any fixed tolerance either buys less speed on some model–task pairs or overshoots the risk budget on others; at a 10% risk budget it deploys a LLaDA2 math configuration that commits over a third more tokens per forward pass, and therefore needs proportionally fewer passes to emit the same sequence. The measured budgets are task-dependent — math clears at a tighter budget than

code [A] [92]. "Cheaper but less accurate" is thereby replaced by a measurable statement: speed is bought at a chosen risk budget. The same selection logic governs the other layers' choice points (objective, architecture, contract tier), which is the subject of Section 10.

The guarantee attaches to a serving configuration. *Engine.* With weights and decoding fixed, the inference engine alone shifts benchmark scores by up to 16.6 percentage points across five engines [121] [E]; in one vision-language family, 4-bit weights cost 1.6 accuracy points and drop the internal error-detection AUROC from 0.95 to 0.80 [51] [E]. Nondeterminism is concentrated in $p \in [0.1, 0.9]$, the cutoff's own neighbourhood [52], and removing it costs more than twice the latency and up to 74% throughput loss [A] [160]. *Precision.* Precision is not even a stable label: vLLM's quantized KV cache defaults to uncalibrated scales of one, and TensorRT-LLM's FP8 recipe differs between SM90 and SM100+ (FP32 against UE8M0 activation scales). Calibration is a deployment artifact that no checkpoint records, and the same weights on a newer GPU are different arithmetic[5]. *Co-tenancy.* Batch composition changes reductions [52] [E], so co-resident requests can change the sampled tokens for the same prompt; the serving stack documents the effect [142] [V]. We treat it as a third determinism term beside task determinism and model stochasticity (serving-path determinism, Section 2.3), so the deployed function is a function of its neighbours. The danger is not that quantization is unsafe — AWQ INT4's safety score stayed within about 1.6 points of full precision on seven of eight models [122] [E] — and the risk sits in the record: a cutoff carries an error rate for the configuration it was fitted on. A stack change is a model change, and the configuration record has to say so.

The decision-model case studies supply the second half of the bridge: systems that return typed probabilities per question (Layer 4) are exactly the systems on which *measured-accuracy-at-coverage* thresholds are computable, and the independent Jev benchmarking line evaluates a commercial decision model on precisely those terms: zero-shot, frozen templates, full rather than sampled splits. Its findings read as a controlled demonstration of this section's principle: the model's choice probabilities measure as well-calibrated and support selective prediction, but its binary probabilities — ranking well as scores — sit poorly placed against a fixed 0.5 default, and thresholds tuned on its training data raise the classification result on a terms-of-service benchmark (micro-F1 from 0.50 to 0.75) [35] [E]. In that benchmark the threshold was the binding constraint on the reported metric. That is evidence about one benchmark, not evidence that thresholds dominate in general. The vendor documentation quoted in the epigraph shows the same stance arriving in industry language, caveats included: cutoffs are fit and validated on held-out data at the option counts the workload uses, and shipped confidence is refitted before gating. Deployment-grade selection, in other words, is a Layer-5 procedure: Layer-0 sets the budget, Layer-4 supplies the scored return, and the procedure picks the operating point.

8.8 The failure envelope of the assurance layer

Every rung above is a component, and components fail. The layer's own envelope deserves the same honesty as the model envelopes of Section 4:

- **Wrong grounding.** The evidence channel can carry stale, conflicting or false documents; grounding makes claims attributable. A system that cites carefully can still be wrong with provenance.
- **Wrong verifier.** Learned checkers have error rates, and one used as a selection signal or

[5] https://docs.vllm.ai/en/latest/features/quantization/quantized_kvcache; https://github.com/NVIDIA/TensorRT-LLM/blob/main/docs/source/features/quantization.md, accessed 2026-10-20; evidence class [V].

reward is subject to being optimized against rather than for the truth (§8.4); verification moves error without deleting it.

- **Correction racing.** A check that completes after the action has committed has no authority. When outputs stream, tools execute, or downstream systems consume immediately, the assurance layer must fit inside the committed-before-irreversible latency budget, which is why reversibility and observability appear as operating-mode modifiers in Section 10.
- **Unsound auditing.** Monitors can be evaded under ordinary task pressure (§8.6); monitor scores validated against proxies can mislead [66]; and a fairness monitor certifies conformance to a specification, not justice. Auditing is only as sound as the quantity it audits.
- **Scoped guarantees.** The theory-class ([T]) bounds are conditional on the calibration distribution; under drift, an unmonitored guarantee is a historical statement.
- **Escalation is also a system.** Human review has capacity limits, fatigue, and its own error rates — detection of naturally occurring errors ran near 24% [49] [E] — so a budget that routes overload to humans has not removed risk, only moved and repriced it (§8.5).

The closing honesty the whole chapter is built to support: *layered defense is a policy decision, not a property.* How many independent checks to stack, which evidence classes to require, and where to set acceptance and escalation budgets are choices made per task against error cost and answer-space openness (Section 10), and they are choices with a ceiling: no layer makes another layer correct. What the Reliability Stack claims for Layer 5 is the honest version of assurance: an error rate you measured, under assumptions you can name, at a residual you chose to own.
The mechanism is sketched in Fig. 15.

9 Case Studies

Seven recurring reliable-AI engineering settings are our case studies. Each is an *instance* of the stack of Section 3, and each is documentation-anchored rather than drawn from a measured population of deployments. What they demonstrate, then, is the framework fitting together, and the seven-row comparison, with Case 6 as the null, tests that. Per our protocol (Section 11), every claim a card rests on is carried in the ledger with its class. Cards mark the class where it changes the reading, and Table 2 gives the per-case summary. Figs. 17 to 23 give the seven architecture sketches; each case's sketch follows its prose.

9.1 Case 1: Retrieval-augmented LLM assistant

The dominant deployment class is an open-ended knowledge task on an autoregressive objective, with provenance deliberately externalized to a retrieval index whose shape — flat, graph, or episodic log — is itself a selection with measured consequences (Section 8.2). Case 1's closed corpus is the instance where flat is the right choice; we keep it and decline the alternative. The reliability literature is dense but disorganized: the trustworthy-RAG surveys collect reliability as per-dimension technique checklists [117, 162], and the hallucination surveys map a single failure mode across a lifecycle rather than a design-objective stack [2, 159]. On the positive side, a live empirical line teaches the model to abstain when retrieved evidence cannot support a response [149] [E], and conformal abstention supplies an assumption-bound procedure for deciding when to refuse [150] [T]. Self-consistency remains the incumbent model-level mitigation [146], with no dedicated survey

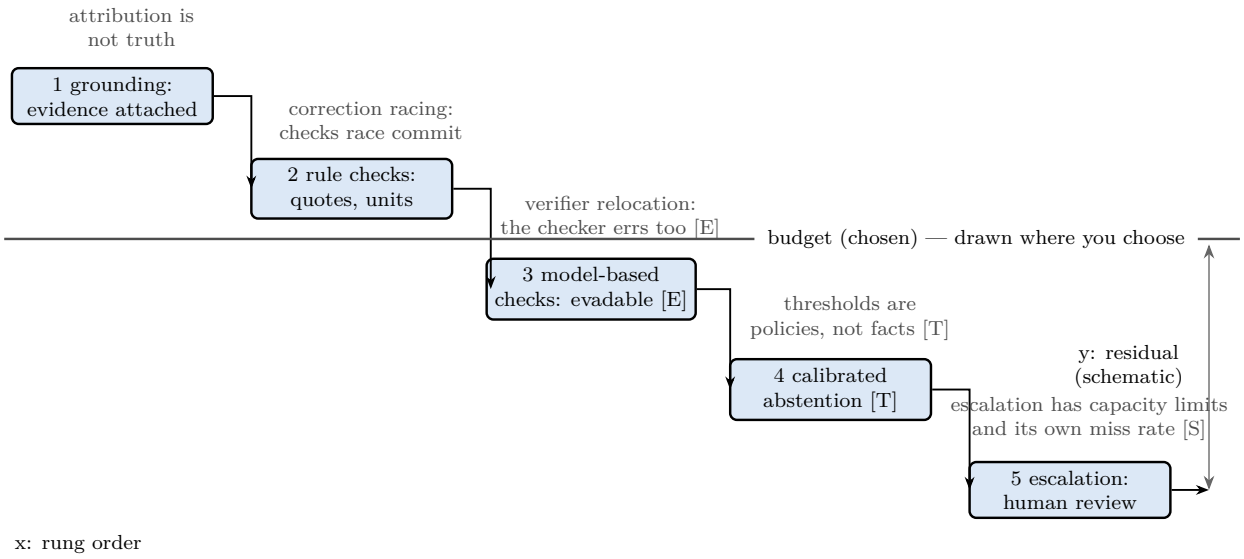


Figure 15: Layer 5 schematic: the chosen budget line against the residual it leaves. Five rungs in cost order (x) against residual risk (y); both axes are ordinal. The residual steps down at each rung and every step leaks — attribution is not truth, correction racing, verifier relocation [E], thresholds are policies [T], escalation capacity [S] — and the gap between the chosen budget and the last step is bracketed as the residual. The fourth envelope item, unsound auditing, is not a rung: monitors are evadable under ordinary task pressure (Section 8.6). No rates are encoded: step heights are deliberately uneven and the crossing point is schematic. The budget line is drawn where the operator chooses — its placement is illustrative.

(a named gap in our ledger). *Inherited failure mode:* hallucination is intrinsic to the objective; compositional effects of chaining retrieval with conformal and self-consistency layers are unmeasured in our evidence, as is human reviewers' catch rate. Fig. 17 marks both gaps as dashed wiring. Remove the L4 contract — citations as structured metadata — and the grounding gate has nothing to check; remove L5 and the contract certifies form only. The gate catches citation mismatch; post-rationalization sits outside it, and Section 8.2 prices that gap at a measured miss rate. The instance is reliable only as that pair.

9.2 Case 2: Decision-model serving

This case exercises every layer. At L0, bounded decisions over closed option sets; at L4 the contract is the typed primitives `choice`/`score`/`noul` (target vocabulary of the Laya-class systems), returned from a single forward pass of a non-generative model [35, 43, 107]. The boundary-setting evaluation literature describes the contract class independently of the vendor [42, 100, 109]: multimodal decision models, per-dataset diagnostic QA gating, calibration metrics. Training aligns with proper scoring rules (log + spherical + ranked-probability score) of the Laya-class recipe; we anchor the recipe to the vendor's shipped documentation [V].[6] The vendor reports threshold-as-policy ("a threshold is a policy you choose from measured accuracy …on your data, not a property of the model"), our L5 epigraph; but the family's reliability claims require the contested-direction caveat: the shipped Typed-Decisions checkpoint was independently measured *under*-confident (signed gap

[6]The vendor-named training scheme shares an acronym with an unrelated older method ("RLCD"); we cite the vendor's usage only, so the label does not ride on the older literature.

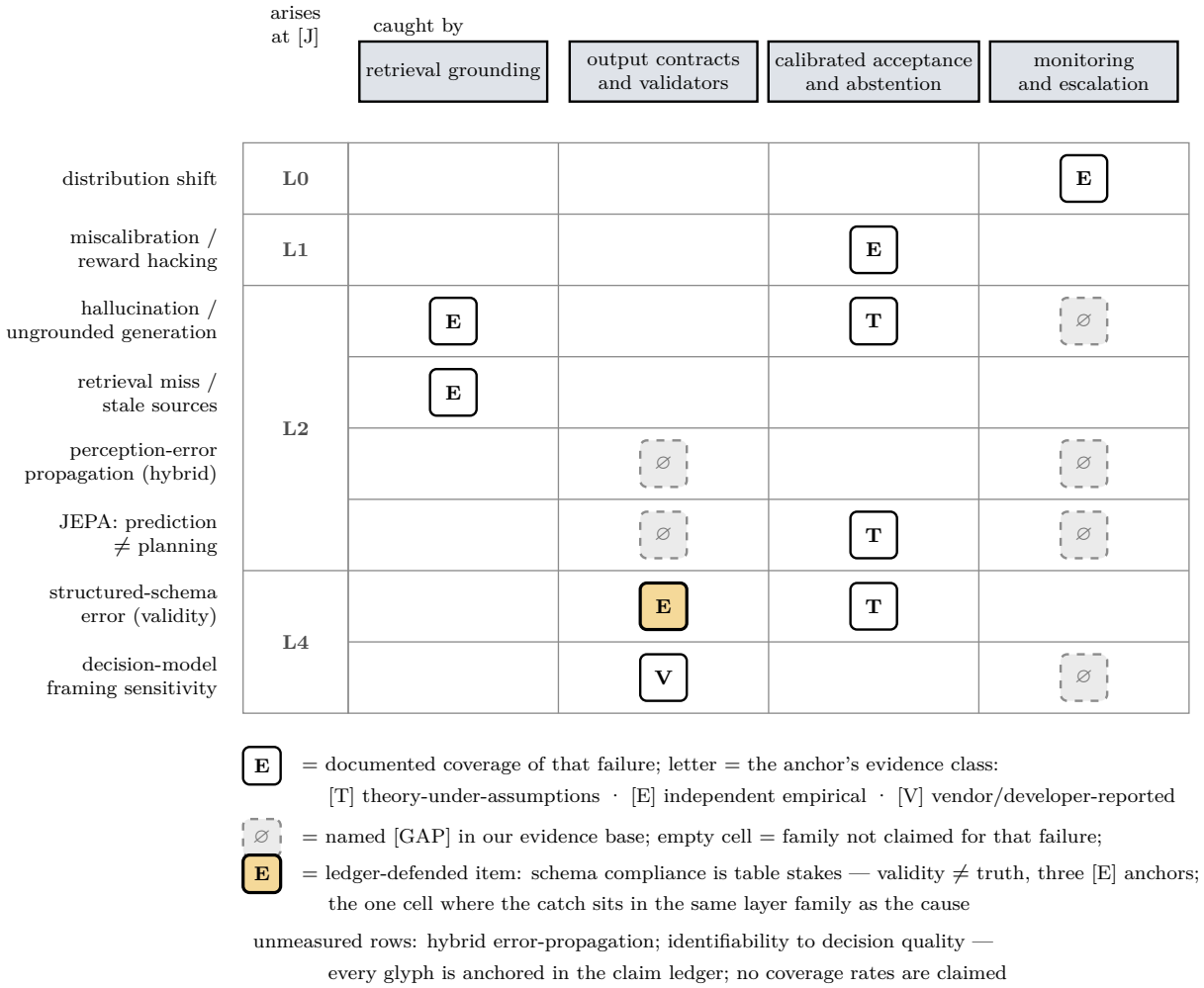


Figure 16: Failure modes sorted by the layer where they arise, against the safeguard families that catch them. Read each row as a sentence — originates at Lx, caught by — and the argument appears: origins cluster at L2 (representation) while filled cells cluster in the last three columns, so the layer where a failure originates is usually not the layer that catches it. The amber cell is the single same-layer catch: structured-schema (validity) error caught by output contracts and validators — and it closes only the validity half; the value error survives (validity ≠ truth). Origin assignment is our analytic cut [J] — distribution shift is the ambiguous row, filed at L0 and stated as such. Glyphs: E/T/V = documented coverage of that failure and the evidence class of the anchor; ∅ = named [GAP]; empty = family not claimed. No coverage rates are claimed.

−0.214) [E], the opposite of the card's framing, and the direction decides which way a confidence-gated cascade fails [114]. The deployed gate — thresholds frozen before evaluation — beats random escalation on both reported tracks, but misses its 10% accepted-error target on each track [E] [114]. The headline 30% of answers flip when option order reverses, and accuracy drops sharply at 50 similar candidates [95] [E]. Third-party entrants now contest benchmark-leader margins with KL/Brier metrics, and those margins carry labelling-teacher saturation caveats [29]. Because a typed decision acts without a human reader in the loop, the verdict is the attack surface: black-box attacks flip the verdicts of verified-reward models without touching their weights [71], and decision

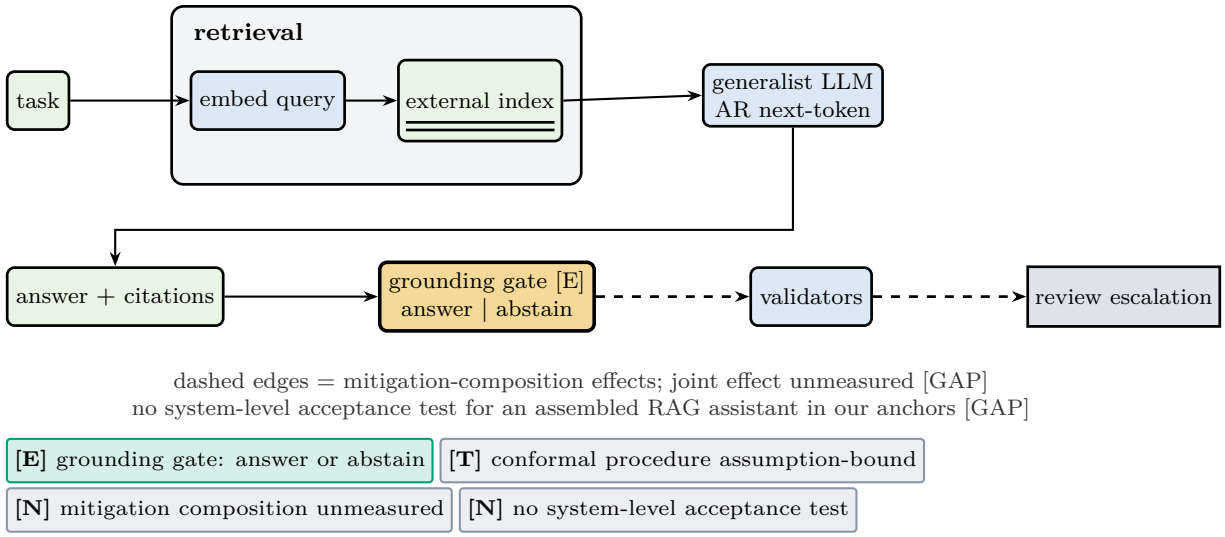


Figure 17: Case 1: retrieval-augmented assistant. Accent = ledger-admissible grounding gate ([E] recovery-abstention line + assumption-bound conformal procedure [T]); dashed = mitigation composition and system-level-acceptance coverage, both named gaps; human review remains unquantifiable [GAP]. Glyph: the two thin rules in the store box = stored records; the external index is a data store that is retrieved from, outside the processing path.

models already serve inside autonomous penetration-testing harnesses [41]. A 2026 red team of a due-diligence deployment broke 25 of 27 Jev configurations at roughly $0.50 per break, its strongest attacker succeeding on 9 of 9 [E] [24]; on that record the typed output did not harden the contract [J].

What this instance leaves unmeasured: no anchor evaluates a fully deployed decision-model system end-to-end (harness + gate + escalation); the joint effect of the documented fixes (temperature refit, option shuffling, shortlist workaround) is unmeasured in our evidence.

9.3 Case 3: Hybrid specialist (Interfaze-class)

This case instantiates an architecture class: task-specific DNNs/CNNs fused into a transformer decoder through a shared embedding space [85] (Fig. 19). Its value is representational: the specialist channel returns structured fields (boxes, coordinates, confidence values) that the stack can check, and the fusion layer absorbs interaction errors when a specialist meets inputs outside its training distribution. Multi-expert adapters have precedents in the literature [22, 134], and the distinction between intra- and system-level routing matters here too [21]. The metadata stream is only verifiable because the specialists return typed fields; a generalist producing the same fields as text would leave the fusion layer nothing to check. Our verdict stays restrained: the case's value is that check. *Inherited failure:* cross-component error propagation is known to be a leading reliability cost in per-component pipelines [128]; all evaluation evidence is vendor-class, and no independent propagation or end-to-end comparison anchor exists in our evidence set.

9.4 Case 4: JEPA world-model planner

This case aims at a conception of reliability that generative objectives cannot reach: latent-predictive training [10], whose planning value is conditional on identifiability theory [38, 87, 158]. The iden-

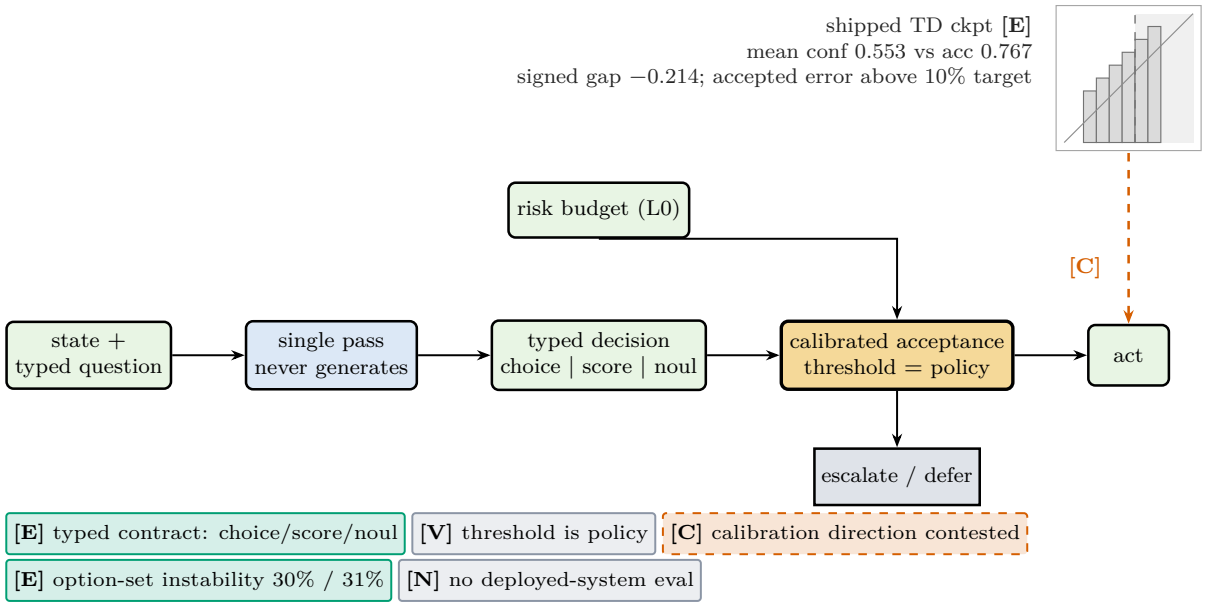
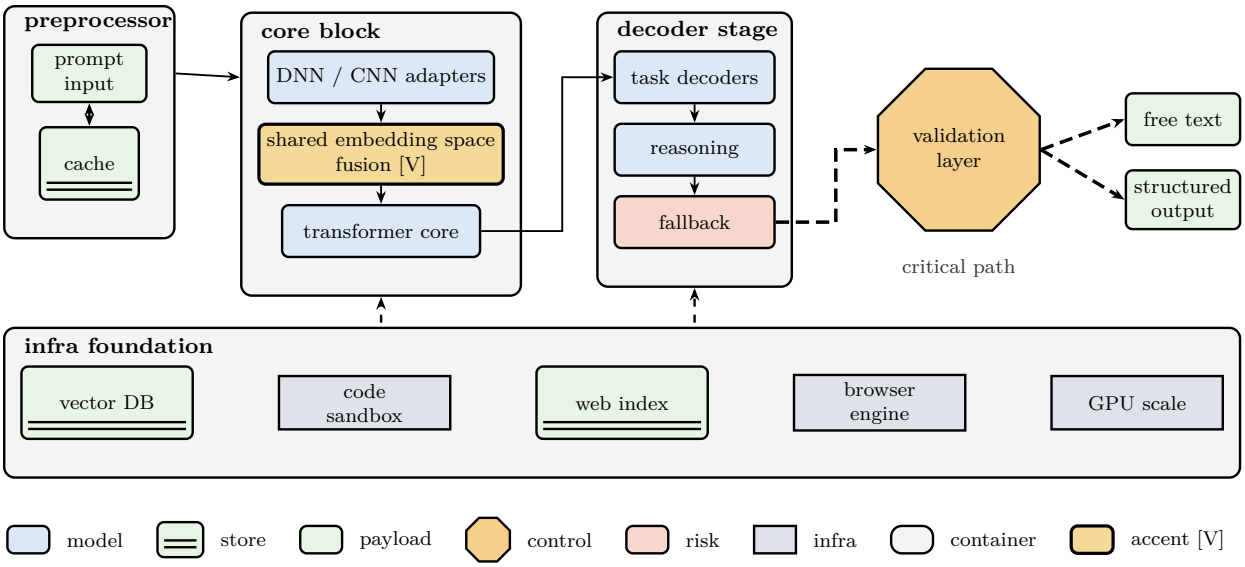


Figure 18: Case 2: decision-model serving. Accent = calibrated acceptance as threshold-as-policy (locked to [V] vendor docs and [E] independent benchmarking). Inset: the shipped TD checkpoint's reliability diagram ([114]) — the bars are the confidence bins, bar height is accuracy in that bin, and the diagonal is perfect calibration, so bars above it are under-confidence; the dashed rule is the policy cut and the shaded band is the accepted set (bin spacing schematic); measured mean confidence 0.553 vs accuracy 0.767 (signed gap −0.214), and the accepted set misses its 10% error target [E]. The vermillion dashed leader marks the [C] claim: the card frames the cascade as failing toward over-confidence, while the independent measurement finds the shipped checkpoint under-confident — the direction is contradicted (anchor named); the measurement stands. End-to-end-system evidence remains a named gap.

tifiability trio supplies only assumption-bound performance guarantees [87, 158]; the transfer-to-language paradox supplies our honest counterpoint (Section 5.2). *Inherited failure:* a systematic evaluation awaiting our full-text read is carried as [W]: the source is cited for a planning-horizon cap and a success-rate fall at 20-step horizons, but its full-text check is incomplete, so those quantities are *not verified* (n/v) here and we do not use them. Text-modality stability remains contested elsewhere in our ledger [36] [E] [C]. Fig. 20 marks the identifiability block as the condition the system must pay for.

9.5 Case 5: Discriminative classifier baseline

The deliberate non-generative control is a mapping head $P(y|x)$ whose L1 foregoes generation entirely. Its reliability machinery carries calibration and selective prediction inventories [61, 103], abstention surveys [148], assumption-bound selective-prediction guarantees [39], and a standing measure of the limit of instrumentation: runtime monitors are a separate component, OOD detection alone does not close the envelope [66], and monitoring itself can be evaded under ordinary task pressure [131]. *Inherited failure:* no anchor in our evidence set benchmarks a plain classifier baseline end-to-end against LLM and decision-model stacks on a shared task; the quadrant limits drawn in Fig. 24 are therefore policy-relevant engineering judgment, not measured comparisons. The same class-probability interface bridges back to the decision-model case (Section 9.2) [95].



[V] vendor-reported architecture (2602.04101); yellow = ledger-defended fusion claim.

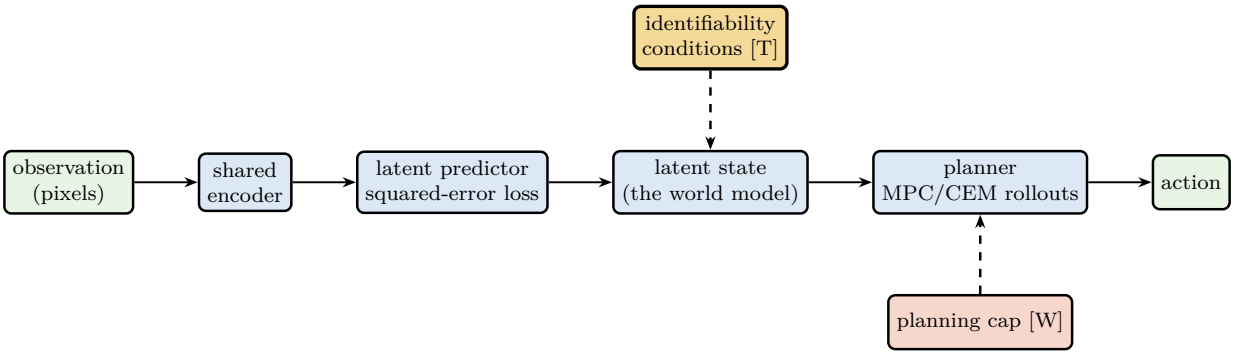
Dashed = vendor-undocumented wiring (validators, monitors, escalation); long dash = reliability-critical path.

Per-component validation; error propagation across the fusion is unmeasured in our evidence set.

[V] vendor-reported fusion architecture | **[V]** per-component validation

[N] error propagation unmeasured | **[N]** validator wiring undocumented

Figure 19: Case 3: hybrid specialist architecture, Interfaze-class [V] (vendor reconstruction; dashed = vendor-undocumented; box with record rules = data store, octagon = control/validation, red = fallback risk). The validation layer is the reliability-critical path.



prediction ≠ planning: the reported planning-horizon cap is unverified [W] — full text pending
identifiability is assumption-bound [T]; no anchor bridges it to decision quality [GAP]

[T] identifiability conditions | **[W]** planning cap persists

[E] text-modality transfer collapses | **[N]** no bridge to decision quality

Figure 20: Case 4: latent-predictive world-model planner. Accent = identifiability conditions [T]; the text layer renders prediction-is-not-planning as a caveat; the planning-horizon evidence is marked n/v in the section text pending a completed verification pass.

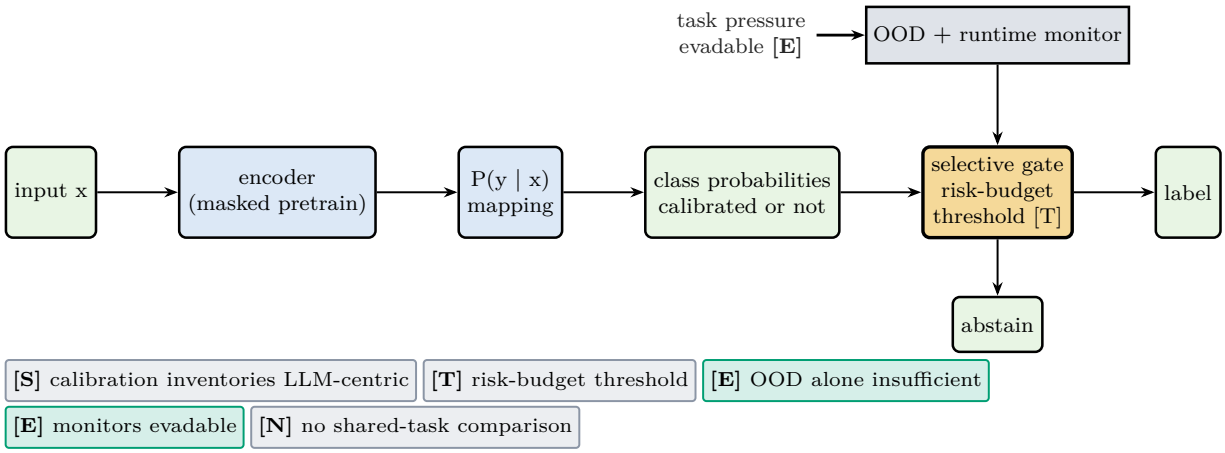


Figure 21: Case 5: discriminative classifier baseline. Accent = risk-budget selective gate [T]; the gate consumes the OOD + runtime monitor, so the monitor's failure is the gate's: ordinary task pressure evades the monitor [E] ([131]), making the assurance layer part of the attack surface rather than above it. OOD detection alone is insufficient [E]; cross-class comparability to LLM and decision-model stacks remains a named gap in the ledger.

9.6 Case 6: Rules / formal-verification pipeline

This is the null-instance discipline case: our evidence base contains no case-study-grade anchor for a production rules-only or formally-verified AI pipeline, and Section 11 records that as an honest null. What we hold instead are three verified enforcement classes: deterministic schema/grammar enforcement at L4 [40, 62, 63], verifier-failure evidence at L5 [127] (validity $\neq$ truth applies to the verification layer itself), and Redline-style risk-budgeted serving-configuration selection [92] (a configuration-selection class adjacent to Case 2's serving lanes, outside the formal loop). *Inherited failure:* statistical guarantee machinery (conformal abstention, provable-alignment, Learn-Then-Test) is uniformly theory-class in our evidence; no independent empirical evaluation of the formal loop over a learned model is in our evidence set.

9.7 Case 7: Production diffusion-LLM serving

Diffusion LLMs have production instances, and the reliability research has not caught up. Mercury 2.5 has been served since 2026-09-08 [76], and the open-weight branch is as concrete: LLaDA2.x ships under Apache-2.0 at 16B and 100B-MoE with its own inference engine and block-parallel decode [20]. Declared and deployed rates diverge: 1,107 tok/s declared for 2.5 [76] [V] against 770 tok/s on a fixed-workload independent instrument and 141 tok/s at router P50 [9, 119] [E]. Two independent instruments differ fivefold on one model, so the vendor figure reads as a ceiling on an undisclosed workload.

The levers split along that hosted/open-weight line. The operating-point procedure of Section 8.7 is executable only where the engine is yours: on LLaDA2 you choose sampler and step count, on the hosted endpoint neither is a request parameter. The current hosted generation (Mercury 3[7]) returns text, usage, and a stream of *revisions* rather than appended tokens, and its request schema carries no seed, no `logprobs` or `top_logprobs`, no n and no step count, with

[7] Mercury 3 was announced 2026-10-19, two days after our 2026-10-17 cutoff. It enters this card as a [V] vendor claim only and is not corpus evidence.

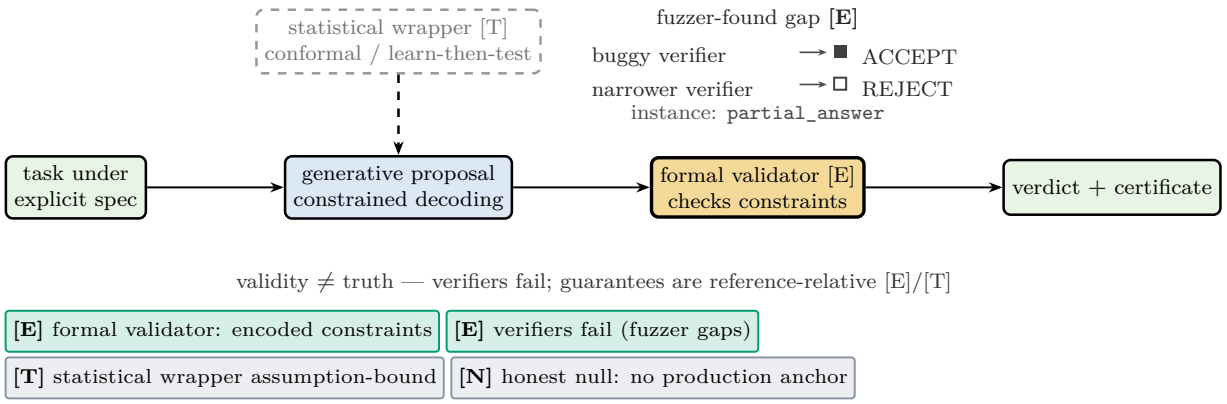


Figure 22: Case 6: rules / formal-verification pipeline. Accent = formal validator [E] (validity relative to encoded constraints). Above it, a paired-verdict trace: the same completion (`partial_answer`) is ACCEPTed under a buggy encoding and REJECTed under a narrower one, so the guard relocates error rather than removing it ([127]) — the narrower verdict is not "correct", only narrower; filled/open markers carry the verdicts, no second colour. This card is a template instance: an honest null, since no production formal-verification anchor exists in our evidence base.

temperature floored at 0.5 [77]. With no per-token score to threshold, the only acceptance surface left is agreement over k whole outputs. The same vendor's typed endpoint is the counter-example: `choice`/`score`/`noul` with probabilities, and ranked 3/135 on an independent decision board [E] [19]. Reliability here is bought at L4, by choosing the endpoint.

What this instance leaves unmeasured: Mercury 3 postdates our cutoff, so no independent quality evaluation of it exists, and for 2.5 the only independent quality signal is a single index score (12, against 21/22/17 on the peers it names), so the hosted generation's quality evidence is thin and mostly vendor-reported. The case concerns the contract and the evidence record. No sampler-variance or NFE study is published either; the inherited failure is sampler sensitivity *made unobservable*, not merely present. Nor does the instance show that parallel refinement buys reliability: latency shape is not an assurance property, and the open-weight branch is the only place where the sampler can be held fixed.

9.8 Cross-case synthesis

Three regularities emerge, and each survives a row-by-row read of Table 2. First, the case cards' *strongest* reliability moves sit above the objective, at L4–L5 for five of seven instances (1, 2, 5, 6, 7) and L2 for the two representational ones (3, 4), rather than at the objective's home layer. That pattern is an argument for cross-layer engineering, and against committing to a single objective family. Second, the last engineering move in each case sits above the mechanism it secures: the decision case needs the L5 gate, and the formal case needs its L5 wrapper re-evaluated. Row 7 is the new information in this table: its reliability move is procurement at L4 — choose the endpoint that reports scores. Third, two of the three families Fig. 25 surfaces as most-contested appear here — calibration direction and latent-prediction stability — so the ledger makes the paper's uncertainty legible rather than uniform.

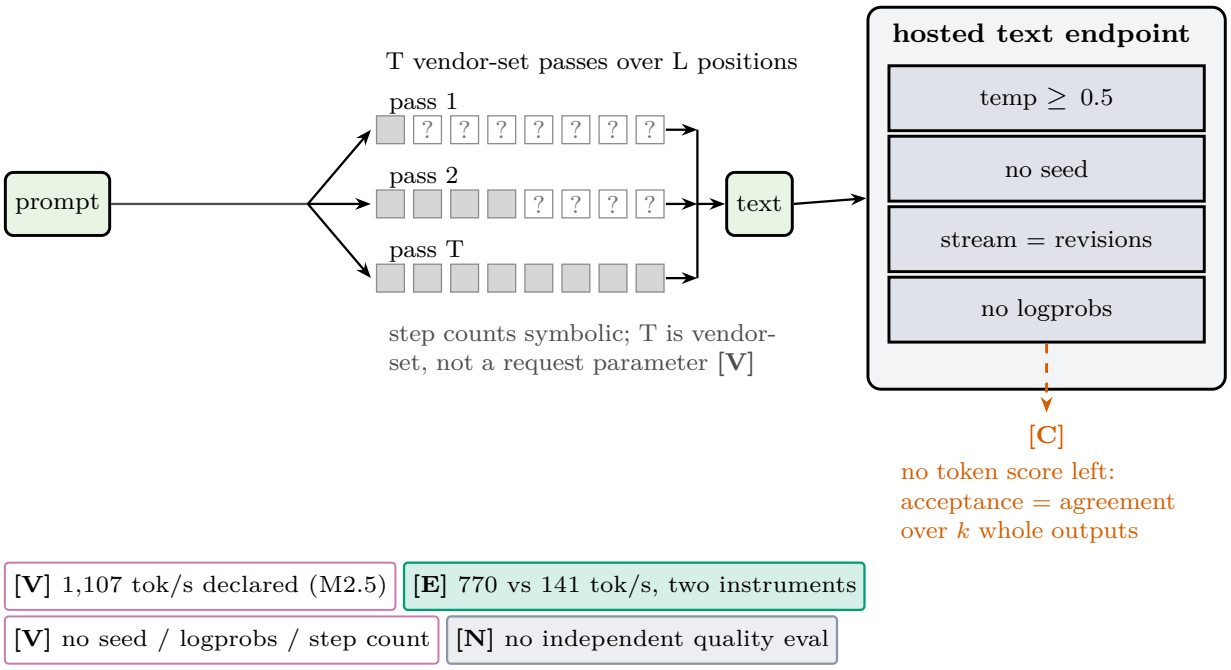


Figure 23: Case 7: production diffusion-LLM serving. Left: L positions refined in parallel over T vendor-set passes — filled = settled, ? = still unresolved, step counts symbolic and no wall-clock drawn. Right: the hosted text endpoint's acceptance surface: temperature floored at 0.5, no seed, no logprobs, a stream of revisions rather than appended tokens. The vermillion dashed leader marks the contested inference $[C]$: with no token score, acceptance reduces to agreement over k whole outputs — the schema silence is $[V]$, the consequence for the L5 gate is ours. Throughput is vendor/independent paired (1,107 tok/s declared vs 770 tok/s on a fixed-workload instrument and 141 tok/s at router P50); Mercury 3 postdates the cutoff, so no independent evaluation of it exists; 2.5 carries one independent index score.

Case	Strongest reliability move	Worst failure the home layer cannot close
1 RAG-LLM assistant	grounding gate + abstention machinery ([E] [149]; [T] [150])	hallucination intrinsic to the AR objective; mitigation composition unmeasured
2 Decision-model serving	typed closed contract + proper-scoring rules ([V] Laya; [E] [35, 109])	interface instability: 30% of answers flip under option reversal [95] [E]; calibration direction contested [114]; gate misses its 10% accepted-error target [E]
3 Hybrid specialist	structured, checkable fields returned by the specialist components ([V] [85])	cross-component error propagation unmeasured (named gap); all evaluation evidence vendor-reported
4 JEPA planner	identifiability-under-assumptions guarantees ([T] [87, 158])	an unverified report of a planning-horizon cap; all quantities n/v ([W] [4]); an independent evaluation reports latent-prediction instability on ambiguous text ([E] [C] [36])
5 Classifier baseline	calibration/abstention/monitoring inventory ([S, T])	monitor evasion under task pressure ([E] [131]); instrumentation needs its own threat model
6 Rules / formal	deterministic validity enforcement ([E]) + distribution-free wrappers ([T])	validity $\neq$ truth: verifiers themselves fail ([E] [127]); honest null on formal loops
7 Diffusion-LLM serving	endpoint choice as the reliability move: the typed endpoint reports scores ([V] docs; [E] rank 3/135 [19]) where the text endpoint reports none	sampler schedule is vendor-side — no step count, seed or token score, so sampler variance cannot be held fixed (named gap)

Table 2: Cross-case comparison. Evidence classes: Section 11. The third column names the mechanism that the instance's own layer cannot close.

10 Task Selection Framework

The stack of Section 3 tells you *what a system is*, and this section tells a practitioner *what to build*. The selection framework routes by task; model preference is not an axis. It is written as a procedure because the assembly is the product. Each of the six steps hands the next a term it cannot invent for itself, and the last one ends at a configuration whose residual risk is named and owed.

10.1 The two axes

The horizontal axis is **answer-space openness**: how far the correct answer set is from enumerable at design time. Well-defined answer spaces support closed contracts (typed decisions, validation); open-ended ones cannot be closed by any contract layer and must rely on grounding and review (Section 9.1). We use *openness of the answer space*. Stochastic inference is a deployment knob; an axis built on it would re-ask the determinism question and leave every reliability limit unaddressed.

The vertical axis is **cost of error**: the severity of a wrong output. Reversible low-stakes outputs allow sampling or iteration strategies; irreversible high-stakes ones push the operating point toward validated acceptance and human escalation, whichever objective family serves the task. Fig. 24 draws the resulting quadrants and the escalation band.

10.2 The procedure

The procedure starts from the paper's one falsifiable cross-layer regularity: failures originate where they are least catchable and are usually caught in the later columns of Fig. 16. Where the checks sit is therefore a decision the procedure makes. It consumes the ledger.

1. **Risk specification.** Fix the three terms that make "reliable" a claim in this paper: the envelope of failure kinds the system must not produce, the budget it must hold, and the assurance path that intervenes as it drifts (Section 2). Mapped onto the framework, those terms are two axes—answer-space openness and error severity—plus two constraints, reversibility and operational budget: three claims, which the framework reads as four terms every later step reads.

2. **Placement.** Locate the task on Fig. 24: how open the answer set is at design time, and what a wrong output costs. The coordinate is an analytic cut [J], the status the map gives its quadrant lists. Record the reason the placement holds rather than treating the quadrant as measured, and keep task openness separate from decoding stochasticity, which is a deployment knob (Section 2).

3. **Modifier routes.** Walk the four placement modifiers — distribution shift, reversibility, observability, operational budget — plus the schedule route, which asks the question the axes never raise: *which factorization does this contract have to inspect?* The schedule is not inherited from the objective; Fig. 11 draws the three factorizations as lanes, and Section 10.3(v) states the route. Each route is stated with its evidence there; any of them can move the operating point without moving the axes, so the placement from Step 2 is provisional until they have been walked.

4. **Probe against the failure matrix.** For each failure kind in the envelope, find the row where it originates in Fig. 16 and ask which later column would catch it; where no column does, the candidate needs another layer or a narrower claim, and the residue is bracketed as the residual of Fig. 15. A candidate that verifies only at the layer where it operates is checking the wrong

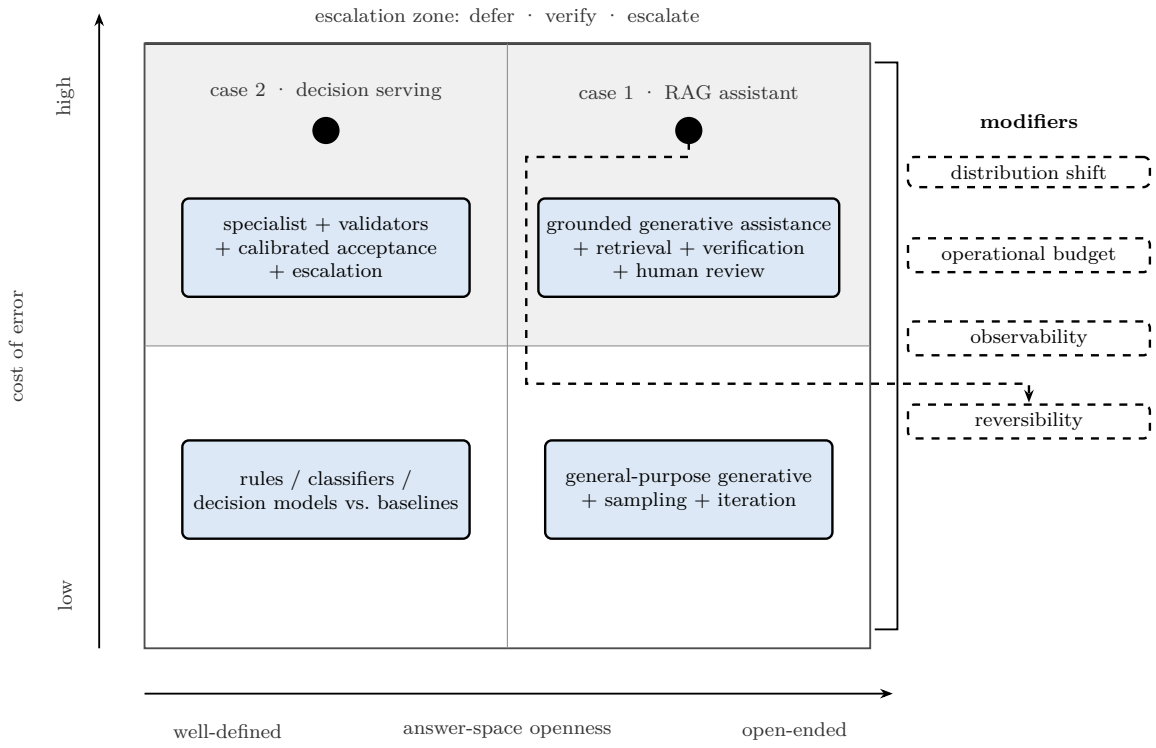


Figure 24: Task-selection map: quadrant mode lists are candidate operating modes [S]; the two probes are our own cases placed illustratively — case 2 sits at high cost because a typed decision acts without a human reader in the loop — and the dashed elbow runs from probe 1 to the reversibility callout — it names the lever that would move the placement, not a measured trajectory; modifier callouts are claims-to-defend; the escalation band is the high-error-cost row (default response: defer, verify, escalate).

place by default: origins cluster at L2 while filled cells cluster in the last three columns. The two probes already on Fig. 24 are our own cases, and they are illustrative: they show what a filled placement looks like before any threshold exists.

5. **Archetype routes.** Read the quadrant's candidate modes and select one to carry: (low cost, well-defined) rules, classifiers and decision models benchmarked against baselines; (high cost, well-defined) specialist plus validators, calibrated acceptance and escalation; (high cost, open-ended) grounded generative assistance with retrieval, verification and human review; (low cost, open-ended) general-purpose generation with sampling and iteration. Two gated variants enter only where their precondition holds: latent-prediction planning under identifiability conditions (Section 9.4), and rules or formal verification where an executable validator exists (Section 9.6). Then read the card of Section 4.2 — what the objective optimizes, which task types it suits, which failure modes remain, which complementary layers cover the residual — and the mechanism sketch for where the design's benefit comes from. The route fixes the objective family and the contract tier; it does not fix the schedule of Step 3.

6. **Budget covenant.** Convert the L0 budget into thresholds and escalation conditions with the existing [T]-class machinery — conformal abstention [150], risk-controlled selective answering [39], and risk-budgeted serving [92] — then fit them on the operator's calibration set, reference

configuration and correctness measure. The cutoff is the cost-based reject option, chosen to hold the error budget on the calibration distribution [59]; never install it as a fixed universal constant (see also the contested-direction caveat [114]); the output is an *operator*, not a "safe zone". Instrument it: monitors, abstention and escalation ladder, plus a scheduled evaluation plan that tracks shift and takeover attempts, the contested claims of Table 2 included.

What the six steps produce is a candidate operating mode with a checked residual-risk story. The framework draws two lines: it treats ranking model classes against each other as a category error (Section 3), and it holds escalation to the residual the evidence supports.

10.3 Modifier routes

Four modifiers move the placement and leave the axes fixed. A fifth route, the schedule, does a different job: it selects which factorization the contract inspects. All five are claims we have to defend, and each answers a question the placement alone cannot settle.

(i) Distribution shift. Widens margins on the reliability side: a well-instrumented classifier degrades unpredictably under shift, making calibrated abstention the floor rather than a bonus [66, 103]. Shift is also the matrix's one ambiguous origin row, filed at L0 and stated as such (Fig. 16).

(ii) Reversibility. Converts some high-error-cost placements back into the iterative regime, because repairable damage can wait for repair. The dashed elbow of Fig. 24 is this move drawn.

(iii) Observability. Determines whether the assurance layer gets signal at all; monitoring under low observability is one of the honest failed rows of Section 9.5, where the monitor is evaded under ordinary task pressure [131].

(iv) Operational budget. Constrains the acceptable cost of the safeguard chain itself; it constrains which configuration is even selectable, not how reliable it should be. A configuration that reaches target quality faster converts its margin into reliability budget (more candidates, more checks) rather than raw throughput; the schedule classes of Fig. 11 make that exchange rate visible, and on XSum the NCM authors spend saved sampling time on minimum-Bayes-risk selection for about 7% higher ROUGE-L within the same sampling budget [A] (Section 8.3).

(v) Schedule. The route that puts the schedule axis into the procedure: it fixes which factorization the contract must inspect, and so where the acceptance surface sits. A schedule is a factorization choice, not an architecture: the same objective admits a left-to-right chain, an iterative denoising sweep, or a coarse-to-fine level order, and what differs across the three is how many sequential decisions a contract must accept and whether any of them can be taken back (Fig. 11, Section 7). The archetype of Step 5 therefore fixes the objective family; the schedule is a separate commitment, and it is the one that decides whether the budget conversion of (iv) buys more passes or more checks. Where a shorter path holds quality, the saved passes are spendable on verification: the hierarchical family reports 21× faster decoding on LM1B at matched parameter scale and 31× on XSum at matched ROUGE-L, at $K \ll L$ [A] [89], and its tokenizer-stage substitutions are what the contract's exactness boundary then has to catch.

10.4 A worked instance: typed decision serving

The six steps are easier to check when one task is carried through them. This instance carries Case 2 (Section 9.2), a non-generative decision model returning `choice/score/noul` from one

forward pass. It adds no measurement of its own; the single cell the paper cannot supply is marked operator-supplied.

1. **Risk specification.** A closed decision over an option set of 50 similar candidates, the size at which the inherited evaluation reports its accuracy drop; a 10% accepted-error budget; no human reader in the loop, because the decision acts. Two failure kinds fill the envelope: interface instability under option order and option-set size, and acceptance miscalibrated against the budget.
2. **Placement.** High cost of error against a well-defined answer space: the quadrant routing to a specialist with validators, calibrated acceptance and escalation (Fig. 24). The placement is an analytic cut [J]: it records that a wrong decision acts and that the option set is enumerable at design time.
3. **Modifier routes.** Shift is presentational here: a paired evaluation holds the inputs fixed, reverses option order, and 30% of answers flip, with accuracy dropping at 50 similar candidates [95] [E]. Reversibility is none once the decision acts, so the gate sits before commitment. Observability is the serving harness's request and decision logs. Budget is one forward pass, leaving no sampling passes to convert into checks. The schedule route does not arise: a single-pass decision has one commitment point.
4. **Probe against the failure matrix.** Interface instability is the decision-model framing row: it originates at L4 and is caught in the contracts column, where shuffling against order and a shortlist against set size both sit. The row's monitoring-and-escalation cell is an empty gap in our evidence, so the gate this instance adds is a design move; no catch is documented. Acceptance miscalibration originates at L1, and its acceptance cell is filled only with a contested direction. The joint effect of the three is unmeasured, and no anchor evaluates harness, gate and escalation together (Section 9.2): the procedure brackets that gap, it does not close it.
5. **Archetype route.** Within that route the family is the decision-oriented one, at the typed-decision contract tier, with the card of Section 4.2 supplying the failure modes it leaves and the complements that cover them. The validators cover form. The contract settles that the answer is a well-formed choice over the offered set and relocates whether the choice is correct.
6. **Budget covenant.** The published gate freezes its thresholds before evaluation and still misses its 10% accepted-error target on both reported tracks, though it beats random escalation [114] [E]. The shipped checkpoint is independently measured under-confident, mean confidence 0.553 against accuracy 0.767, signed gap -0.214 [114] [E]. That sign contradicts the card's over-confidence framing and decides which way a confidence-gated cascade fails, so the operator refits the cut on its own calibration split a threshold is a policy chosen from measured accuracy on the operator's own data, not a property of the model [V]. Coverage at the 10% budget is not reported in this paper; that cell is operator-supplied. Escalation: any accepted set that misses the budget routes to human review, with a measured catch rate. Owner: the operator. Re-evaluation trigger: any checkpoint change or execution-configuration change, or any option set above 50.

Shuffling, a shortlist and a recalibrated gate have not been composed end to end. That residue is why the instance stops short of a guarantee and hands back a configuration with a named gap and an owner for it.

10.5 Why this is useful beyond standard practice

Existing deployed reliability practice solves *per-dimension* problems [117, 159, 162]: a definition each for calibration, RAG honesty, or content quality; each works, is checkable, has failure modes; assembling them for a concrete task is what breaks. Our framing explicitly conditions on L0 and treats the assembly as the product: each safeguard layer declares what it covers, what its unverified residues are (Fig. 16), and which verified substitutes exist if a residue is deal-breaking. The matrix narrows the candidates to the verification-anchored set it enumerates, each with its ledger record, and the evidence map (Fig. 25) tells the reader which of these claims are contested.

The regularity can be tested outside our own matrix. We coded four incidents chosen for their public record rather than their fit—Amazon's recruiting tool (2018), the Dutch childcare-benefits risk model (2013–2021), Zillow Offers (2021), and the Air Canada chatbot (2024)—with the coding rule fixed before the cases were assigned: origin is the earliest layer whose documented design choice made the harm likely, and a catch is a safeguard inside the deployed system that fired before harm [5, 32, 108, 163]. In all four the earliest documented origin is L1 or L2, and none was caught at or before that layer: three had no internal catch at all, and the fourth was stopped only by project cancellation at L5. Read at the strength the sample supports, origins cluster early; the sample is too small to locate the mode within L1–L2. The later columns describe safeguards that exist rather than safeguards that fire. Zillow's documented L4 co-origin is the boundary case, and counting nominal review steps as catches would inflate the catch side.[8] The coding manual and every assignment are Appendix A (Table 4).

11 Evidence Typology and Review Protocol

Our claim standard replaces adjectives with ledger entries. Every reliability claim we make about a method or system carries one of the following classes, applied at the level of individual numbers, not whole papers.

11.1 Evidence classes

[T] Theoretical promise under explicit assumptions (conformal abstention, provable alignment, identifiability results, distribution-free risk bounds). Such a claim carries its assumption set forward, and the guarantee is conditional by construction.

[E] Independent empirical: evaluation of an artifact whose authors are not the evaluators, whether the evaluators are a third party measuring what someone else built, shipped or not, or our own re-verification of reported results against released artifacts.

[V] Vendor-reported: developer-documented claims from a model card, engineering note, or commercial system description. Legitimate evidence of what a deployed system does; not evidence that an unstudied analogous mechanism would transfer.

[A] Author-reported: results the artifact's authors report, whether or not they ran the experiment (preprint tables, released evaluation-harness runs, self-evaluations of their own prompt or method). Such a report documents what that team demonstrated; it is never independent reproduction, and where artifacts are released we record that verification is checkable, not that it happened.

[8] One coder, coding rule fixed before assignment, not blinded to the matrix; a retrospective coding, not a benchmark.

Token	One-line meaning	Test a reader applies
[T]	theory under stated assumptions	does the claim carry its assumptions, and would the guarantee fail without them?
[E]	independent evaluation of someone else's artifact	did evaluators who did not author the artifact run the measurement?
[A]	results reported by the artifact's authors	is the measurement the authors' own report of their own artifact?
[V]	vendor or developer documentation	is the source the vendor's own description of a deployed system?
[S]	synthesis verified against the primary records	did we check the aggregation against those records?
[W]	watch/pending: verification incomplete	is the check still incomplete?
[C]	contradiction with a named anchor	is the contradicting anchor named?
[N]	named gap: a scoped, still-open absence	is the absence scoped to what was searched?
[GAP]	the specific unmeasured link behind a gap	which step has no anchor?
[J]	our analytic judgement; no chip	is this cut ours rather than a verified aggregation?

Table 3: The canonical evidence alphabet: six classes ([T], [E], [A], [V], [S], [W]) and four annotations ([C], [N], [GAP], [J]). This is the single alphabet every other surface cites; no surface defines a second one.

[S] Survey or synthesis where the aggregation is verified by us against the primary records, or where our own citation-graph verification substitutes for reproduction. Our own analytic cuts are [J], distinct from [S]: a placement or ordering we choose is judgement over the record.

[W] Watch/pending: entities real enough to have acquired separate documentation while our verification remains incomplete, or a specific result whose full-text check is not yet complete (the card says which). We keep such entries explicit in the ledger so a reader can tell what is solid from what is merely plausible.

Other surfaces list only the classes they use and defer to Table 3.

Separately from these classes, four annotations record what a card or caption is reacting to, while the class carries how strong the evidence is. [C] marks a specific contradiction (the anchor is named); [N] is an absence flag — a scoped, still-open gap in what has been measured, not a negative finding; [GAP] marks the specific unmeasured link behind such a gap, used where a card or a table points at the step that has no anchor; and [J] marks our analytic judgement, used in figure captions, figure annotations, and procedure steps where a cut is ours rather than a verified aggregation. Where a claim carries two classes we write them [T, E]: both apply, and the first listed is dominant. A bare tag placed directly after a citation is also a valid citation-class badge: it classifies the cited result.

No evidence class admits a guarantee of universal correctness: [T] names assumption conditions, [E] measures one deployment configuration, [V] reports one vendor's own design.

The classes encode provenance, not validity: a tag records who ran the measurement, never whether the instrument measured the construct the citing sentence names. An [E] row can rest on a contaminated benchmark, on a harness whose prompt format supplies half the effect, or on a delta inside sampling noise, independently attributed and still not evidence for the claim above it

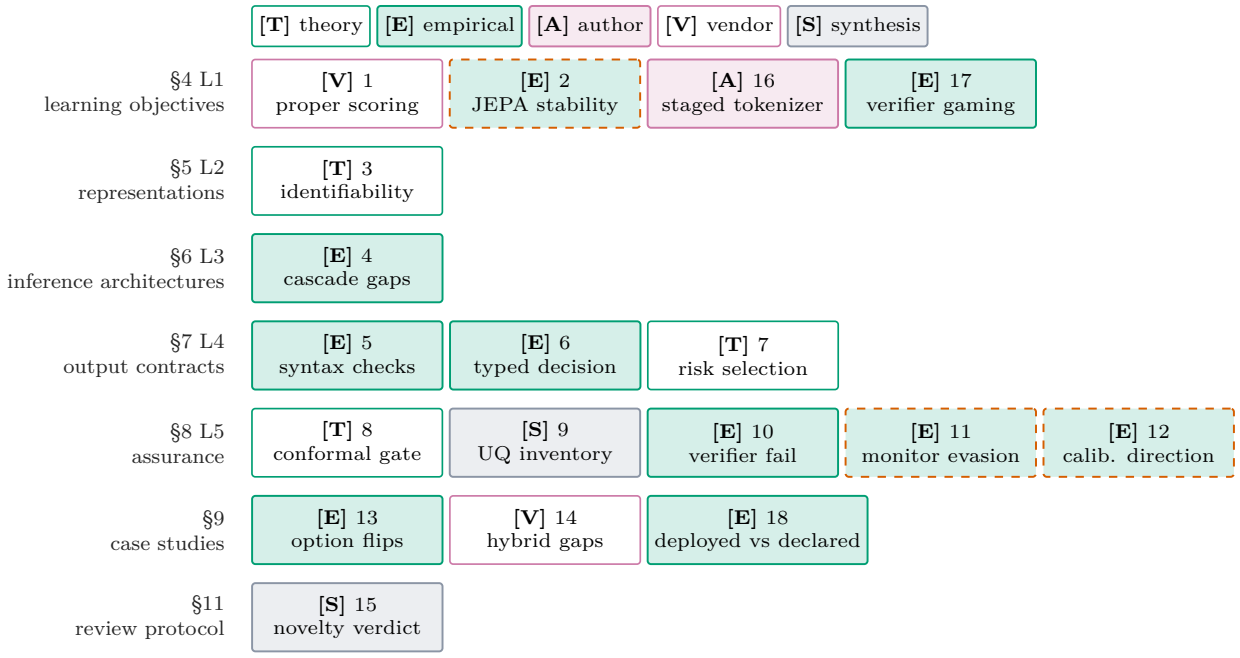


Figure 25: Evidence ledger: one chip = one claim family from the claim ledger; the row is its paper section and the chip order inside a row is ledger order (both our analytic judgement [J]). Row length carries how many families that section contributes: one chip per family; no chip's width and no row's white space encodes a quantity of evidence. Class is carried by the chip: [T] outline, [E] filled green, [A] filled pink, [V] outline pink, [S] filled slate; chip class is the dominant class of the single claim it names, while a Table 1 row tag marks the strongest anchor of the family. The full alphabet, including [W] and the annotations, is Table 3. Contested stays an annotation: the same class chip, dashed vermillion border, for the three most contested families (2 JEPA-in-language stability, 11 monitor-evasion attackability, 12 calibration direction) — the [C] annotation, surfaced rather than hidden. No family in the ledger is [N]; thin rows are counts, not findings. No rates, sample sizes or per-paper counts are encoded.

[1, 16, 27, 83]. Leaderboard position is the same exposure one level up, where rank rewards access to private variants rather than reliability [136]. Where an anchor reports no interval we print its number as the source printed it and do not read a point estimate as a difference [23, 110].

11.2 Claim-ledger construction

For every claim that a card surfaces, the ledger records anchor (ID or artifact), verbatim-vs-paraphrase status, the numbers as written in their original context, the direction (stats, benchmark structure) with the verification date, and the layer(s) the claim touches. Any claim cited from sampling memory fails our protocol; every anchor is re-fetched live (arXiv API verification with Microsoft/Elsevier content checks where applicable) before use in this manuscript.

11.3 Exclusions register

We exclude claims from the ledger by a single rule: no claimworthy statement enters unless a primary source stands behind it. The exclusions this cycle are concrete: a production rules-only / formally-verified pipeline (Case 6 is carried as an honest null on this very point); a JEPA-"SWM-distractors" benchmark claim that failed verification and was never admitted to the ledger. The list continues with a structured-hallucination benchmark without recoverable verification, a universal >20-option cost rule stated as a law (the vendor's per-system mechanism quote remains a [V] row in its own section), the 0.85 "default" confidence threshold (replaced by threshold-as-policy), and a coinage for which we could find no existing use. We treat it as ours to coin only if the definition appears in our own terms. Each excluded row is retained verbatim in the build's review ledger so the exclusion itself is auditable.

11.4 Reproducibility record

All searches ran the arXiv API (id-list/title/abs search, ≤ 10 IDs per request) plus publisher DOI metadata lookup, with snapshots per batch in the project's scan directory. The paper's claim-to-evidence map is Fig. 25, and every claim marked contested in this manuscript gets its caveat in the same section. Search cutoffs: 2026-10-17/18.

12 Related Work

We position against the three verified classes of prior work: model catalogues, failure-mode/deployment taxonomies, and per-axis surveys that overlap one or two of our layers. Where we code a competitor, each of the five positioning axes is *treated* (the work devotes a section or a sustained passage to it), *partial* (it gets a paragraph or a single design choice), or *absent* (neither); the coding is categorical, and no axis takes a numeric score.

12.1 Catalogue genre: model-family surveys

The closest catalogue is Almuammar et al.'s *Generative Artificial Intelligence Models: A Survey* [3], which classifies generative model families, their training signals and benchmark behaviors across 36 pages. We verified its full text. On our five positioning axes it is partial on three — learning-objective contrast, reliability, and selection — and absent on the other two, decision interfaces and the integrated stack. The objective contrast is weak, every "reliability" mention is an adjective, and the work is a catalogue rather than a selection framework; contract, validity, routing, and decision-model mechanics do not appear. A second survey covers the same genre on arXiv [155] and spans diffusion and real-world development; the most complete diffusion survey consolidates an adjacent space [97, 152]. We cite these as background figures, and the distinction is explicit: their genre is *what generative models exist*; ours is *which stack instance to select for a reliability-constrained workload*, a question they define but do not answer.

12.2 Failure-mode and deployment taxonomies

The closest overall competitor is the MDPI survey of Ricciardi Celsi and McCann [128]: standardized deployment patterns, an AR-large-language-model failure taxonomy, a claims-to-evidence map, and a JEPA/H-JEPA section. Our verified comparison finds it strong on assurance and assessment structure — a mean of 3.4/5 on its own nine-row coverage table, which is that survey's rubric rather than our five-axis coding — but with *zero* coverage of the decision-model family

(grep-verified: no decision-model vocabulary appears), no learning-objective taxonomy, and only partial coverage of the routing axis: its selection apparatus is one binary architectural decision (token-centric vs. adding a world-model component), scoped to AR pipeline deployment. It tells a deployment engineer what to check after their single architecture choice; our framework selects across stack instances, L0 first. The RAIE survey of test-time scaling for agentic LLMs (An et al., IJCAI-26 survey track, [6]) is the closest survey-level competitor to our task-selection map (Fig. 24). It covers only the inference-time part of our Layers 3/5 (its four scaling dimensions are reasoning, acting, interaction, evolution), and it offers a decision-tree selection guide scoped to agentic test-time strategies; learning objectives, representations, output contracts, decision models and cost-side routing are out of its scope, and its roadmap draws no evidence-quality distinction. A January 2026 five-axis survey of 147 agentic systems [37] goes the other way, auditing the evidence behind architecture, memory, planning, coordination and safety — an assurance map of the field.

The objective space itself is still expanding while this survey is drafted: Next-Concept Modeling (Kumbong et al./Hazy Research) adds a coarse-to-fine next-scale code-hierarchy generation family between the generative rows of Table 1 [89], a fresh instance of our Table-1 pattern in which new objectives move inherited failure modes rather than removing them. The lineage is explicit: NCM carries next-scale prediction from visual autoregressive modeling [138] into text and inherits the concept-level ambition of Large Concept Models [90] and Mimir, a 1.6B concept model whose pretraining corpus spans 46 languages with instruction tuning in 35 [113]. Where the hierarchy comes from is the question NCM leaves open: its K levels are fixed by a frozen tokenizer, against the learned chunking of H-Net [75]. The nearest same-objective competitor, NCP-ArchPreview, has targeted coarse-to-fine latent text generation since September 2026 and does not appear in NCM's own related work [78]; no survey in our verified matrix places the two together.

12.3 Per-axis surveys on our layers

Trustworthy-RAG surveys treat reliability as dimension checklists inside retrieval pipelines [117, 162], and hallucination surveys track a single failure mode across the model lifecycle [2, 159]. UQ and abstention work makes calibration the reliability question [61, 103, 148]; world-model surveys cover prediction and its deployment in robotics [99]. Routing and cascade work began as cost allocation [26, 81, 105] and now carries reliability: Gatekeeper tunes a small model's confidence so that deferral is a calibrated decision rather than a cost threshold [123], a cascade that ends in a human expert rather than a larger model turns its last rung into an accuracy decision [48], and the routing characterization these extend [17, 135] is a decision interface as well as a price. MoE reliability is benchmarked within one instrument [21, 25], and looped-depth on unit tasks [96]. Four bodies of work come closest. The full-stack-safety survey [145] maps data/training/deployment threats but is a roadmap of threats; the HuggingGPT- and Chameleon-class agent surveys [22, 134] describe orchestration patterns without selecting among them. The IEEE CAI 2026 system-level failure-mode taxonomy [141] catalogues fifteen hidden failure modes of LLM applications with no routing decision attached, and Select-then-Route [132] is the closest published analogue to C2 — taxonomy, route, cascade, escalate — except that its taxonomy partitions query classes.

Each covers a *slice* of our frame; no survey among the works we examine here (Section 11) integrates the five positioning axes (learning-objective taxonomy, decision interfaces, verification/assurance, actionable selection framework, integrated L0–L5 stack) into one engineer-facing selection procedure — the gap Section 10 addresses. The thesis has independent support: Rabanser et al. [124] measure twelve reliability metrics along four dimensions across fifteen models and find that capability gains buy only small reliability gains.

12.4 What we claim

We claim no single fixed best method, no new objective, and no novel mechanism. The contribution is the cross-layer link: which stack instance to select, tagged with the class of evidence behind each claim, so that each layer's reliability consequences are auditable. The claim-ledger discipline (Section 11) is itself part of the contribution: contested rows are surfaced (including in Fig. 25), and excluded claims are listed with their reasons.

13 Discussion and Open Questions

13.1 Where the stack itself fails

The honest reading of our own frame is that cross-layer engineering buys no reliability without instrumentation, and that two failure modes recur across the case set and bound what the frame can promise. First, per-dimension certification does not compose: a validated contract plus a calibrated gate plus a grounded retriever can still fail on their *interaction*, and no anchor in our evidence base measures that composition (cases 1 and 3; Section 9.1, Section 9.3). Second, verification mechanisms are themselves attackable or miscalibrated [127, 131] (cases 6 and 5), so the assurance layer needs a threat model of its own.

13.2 What the field still lacks

Eight gaps remain open, and each is phrased as a testable ask: (i) *end-to-end system evaluations* of assembled stacks (harness + gate + escalation) under adversarial conditions; (ii) *certified composition* results for chained safeguards (conformal ∘ validators ∘ grounding), which extend the [T]-tier machinery to mismatches between layers; (iii) *propagation benchmarks* for hybrid specialist stacks, where the Case 3 gap (Section 9.3) is the cleanest measurement target our evidence set exposes; (iv) *language-native latent-prediction training signals* that survive the conditional-concentration objection (JEPA paradox), with assumptions actually testable; (v) *decision-model interface stability*: order robustness and option-set scaling are measurable, and current evidence suggests they are not incidental [95]; (vi) *reconstruction-fidelity measurement for staged-tokenizer generators*: a test that separates token-exactness from meaning-equivalence, so a contract can state which one it enforces; (vii) *escalation-queue measurement*: precision, recall and minutes per review for reviewers working a production LLM escalation queue, which no study in our sweep reports (Section 8.5); (viii) *a configuration record for deployment*: the engine, precision and co-tenancy a score was measured under, published with the cutoff fitted on it (Section 8.7).

13.3 What we refuse to compare

Layer identities are inherited envelopes, not benchmark scores; we decline direct comparisons across families (e.g., discriminative vs. generative "hallucination rates") when the error currencies are not measurable on a shared task. Where the evidence base has no shared-instrument anchor we say so, and where it has one, each row names the instrument it rests on rather than pooling across instruments (Table 2).

13.4 Limitations

Scope is LLM-centric (text and typed-decision workloads); multimodal and embodied control appear only as instances of an architecture. The paper is survey-grade: every claim either resolves

to a public artifact we checked or is tagged where checking stops — [W] pending, [N]/[GAP] no anchor — and [V]/[A] mark evidence that is documented but not independently reproduced. We run no new benchmarks; contested rows in Fig. 25 remain contested at submission time and are labeled as such. The seven case studies cover the stack's decision space but not exhaustively; the exclusion register applies to our choices as it did to the candidates'.

The three validity bounds — instrument, contamination, precision — are fixed in Section 11, and they apply to our own anchors as much as to the sources we cite. What is ours to disclose is narrower: our empirical numbers are point estimates from single harnesses, and a point estimate is not a difference [23, 110].

The boundary to standards and regulation is a division of labor, not a crosswalk. The NIST AI RMF 1.0 [115] (January 2023, voluntary, and under revision as of this writing) and its Generative AI Profile (NIST AI 600-1 [116], July 2024) specify risk-governance functions: govern, map, measure, manage. ISO/IEC 42001:2023 [80] defines a certifiable AI management system. The EU AI Act (Regulation (EU) 2024/1689 [46]) specifies provider and deployer duties; the digital-omnibus Regulation (EU) 2026/1744 [47] defers the Chapter III high-risk obligations for Annex III systems to December 2027 (Annex I systems to August 2028), while the post-market monitoring and incident reporting duties (Articles 72–73) stay on the 2026 timetable. None of these instruments rules on which layer absorbs a failure; their unit is the organization, not the stack instance. They prescribe process: who must document, assess, monitor, and report. A certificate attests to the process, never to the residual error. Our five axes ask a different question: which stack instance to select for a reliability-constrained workload. A management-system certificate does not answer it. The relation runs one way. Our evidence classes can describe what a conformity assessment rests on, and we do not translate controls into layers or claim that any certificate implies a reliability property. A reader who needs compliance should read the instrument; a reader who needs a selection decision should read Section 10 and then the instrument.

13.5 Four live fronts

Four fronts are live, and we do not forecast which matures. Our claim is narrower: each would change a named layer's contract, and none would replace the routing thesis that reliability is assembled across L0–L5 and is not won inside any single model. Coarse-to-fine text generation [78, 89] is a Layer 1 objective family with a Layer 4 consequence: its level order is a schedule (Fig. 11) and the frozen tokenizer behind it is the acceptance surface; the authors' own 97M→725M check still shows scale helping inside the family [A]. Whether that hierarchy is learned, as in H-Net's dynamic chunking [75], or fixed by a tokenizer remains an open schedule question; the atlas's three lanes differ in commitment count and revisability, and learned chunking would move both.

Verifier-driven test-time training [112] closes a loop the stack now draws open: adaptation at inference is only as reliable as the verifier choosing its pseudo-labels, so a Layer 3 mechanism imports a Layer 5 obligation.

Risk-controlled serving [92] turns the L0 budget into an operating point rather than a fixed constant: the budget rule of Section 10 made executable, where same-quality-faster buys candidates and checks rather than raw throughput.

Production diffusion serving [20, 77] is the fourth. A hosted text endpoint refines blocks in parallel over a number of passes the vendor sets, exposes no step count, seed or token score, and floors temperature at 0.5, so the contract question becomes which endpoint reports a score at all; the open-weight branch keeps the sampler configurable. Mercury 3 postdates our cutoff, so we read it as vendor documentation [V], not corpus evidence.

13.6 Conclusion

We asked when scaling a general-purpose model is the right lever and when a changed learning objective, representation, decision interface, or verification pipeline is a better route to reliable AI (Section 1). The evidence supports a narrower answer: both levers are real, but they buy different currencies, and no currency is guaranteed by any single layer. Objective families fail through design choices (an envelope they inherit), and the scaling story that dominates current practice sells fluency and coverage without the contract, calibration and assurance machinery that makes those properties deployable. Our contribution is a cross-layer taxonomy (L0–L5) that names the decision point for each reliability claim, and a selection framework that routes by task rather than by model preference. The claim ledger separates theoretical guarantees, independent measurement, author- and vendor-reported results, synthesis, and what remains genuinely contested, so that the next reliability argument starts from an evidence map rather than a benchmark leaderboard [136]. Seven instances, one of them an honest null, showed the same shape: the reliability-bearing move sat above the layer where the failure originated, and the newest one, diffusion serving, moved by procurement at L4 rather than by engineering above the sampler. Configuration-relative reliability changes what a cutoff means: an error rate belongs to the engine, precision and co-tenancy it was measured under, so a stack change is a model change and the record has to say so. The residual is a quantity the operator measures and owns.

A Coding manual for the retrospective probe

This appendix publishes the manual behind the probe in Section 10. It is written to be re-run: every row of Table 4 records the layer, the documented design choice that puts the origin there, and the safeguard inside the deployed system that produced, or failed to produce, a signal before harm. A reader who disagrees with an assignment can see which cell to disagree with, and the manual is what fixes it.

Two limits belong on the record. The rule was fixed before the incidents were assigned, the cases were chosen for the quality of their public record rather than their fit, and the coding is retrospective: one coder, unblinded to the matrix, with no agreement coefficient to report because there is no second coder. And the count is four. "In all four" is a statement about these cases; nothing in Section 10 should be read as an estimate of how often failures originate early. Record quality is also what makes design choices visible, so the probe tests applicability, not generalization.

Three rules fix the coding. **Origin rule.** The earliest layer whose documented design choice made the harmful outcome likely — not the layer where the loss was recorded (finance, liability, press), and not the layer where a fix was later attempted. **Catch rule.** A safeguard *inside* the deployed system or its pre-deployment gate that produced a blocking or escalating signal *before* harm; external actors — courts, regulators, journalists, plaintiffs, auditors — are recorded as *none*, because the matrix's columns are internal safeguards. **Ties and selection.** Where two layers are jointly causal the earliest is recorded and the co-origin is named in the last column; cases were chosen for record quality, not fit.

Incident (year)	What failed	Origin: layer and documented choice	Catch: layer and signal	Alternative coding
1. Amazon recruiting (2018) [32]	A résumé ranker trained on a decade of mostly male applications learned to penalise "women's" and two women's colleges.	**L1** — the target was similarity to past hires; the historical label carried the disparity.	**L5** — internal review and cancellation before general deployment; the same-layer patch (neutralise the term) did not hold.	**L0** is defensible: "top talent" was never specified, so no objective could have been right; the record names the label.
2. Dutch childcare (2013–21) [5]	A risk classifier used nationality as an input feature; non-Dutch families were scored higher, investigated, and had benefits clawed back.	**L2** — *Dutch citizenship: yes/no* as a feature; the report also records a self-learning element.	**none** — the review step carried no reason code; exposure came from the press, the data-protection authority and the courts.	**L0** if the risk definition is the seat (fraud risk including nationality); we code the documented artefact, the feature set.
3. Zillow Offers (2021) [163]	An iBuying unit made cash offers from a model built and measured for current market value, then absorbed the gap when resale values fell.	**L1** — the objective and error metric priced value today (A) while leaving resale value later (B) unpriced; the loss is the $A - B$ gap.	**none** — no internal check priced forecast error before offers; the first signal was the quarterly write-down.	**L4** co-origin: the estimate *was* the offer and experts were barred from adjusting it; under that reading one of the four origins moves to L4. Weakest of the four.
4. Air Canada chatbot (2024) [108]	The chatbot asserted a retroactive bereavement fare that contradicted the airline's own published policy.	**L2** — the answer was generated with no anchor to the authoritative policy text.	**none** — no check against the policy page; the claim surfaced only when the airline refused it.	**L4** if the licence to answer policy questions definitively (no citation or validity contract) is the seat.
Excluded: named, left uncoded — the record does not settle the origin, so coding would record a preference.				
Robodebt (2016–2020)	Income averaging by a rules engine, with no learned component; debts were raised from a mismatch between reported and assessed income. Exercises L0 and a rules layer the paper does not model; useful, and outside the failure class the matrix is about.			
Detroit facial recognition (2020)	A recognition model's candidate list was treated as an identification. The record does not settle whether the defect is the recognition model (L2) or the decision rule that treats a candidate list as an identification (L4).			

Table 4: Coding manual for the retrospective probe: the origin and catch rules, the four coded incidents, and the two excluded cases. Each assignment records the documented choice behind it, so it can be re-derived from the table; the layer boundaries are our analytic cut [J], and no agreement coefficient is reported (one coder).

Notes. In all four coded cases the earliest documented origin is L1 or L2 (two at L1, two at L2), and no internal safeguard produced a blocking or escalating signal at or before that layer: three had none at all, and the fourth was stopped only by project cancellation at L5. No case in the sample carries an L0 origin. A nominal review step that carries no reason code is not a catch (Dutch case): the catch column describes safeguards that exist, not safeguards that fire. Zillow's L4 co-origin is the boundary case and the weakest of the four codings; counting it at L4 lowers the early-origin count to three. One coder, rule fixed before assignment; retrospective coding of public records.

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